



LX100/200

Hints, Tips and Tricks

A compilation of commonly used procedures and diagrams to aid the Field Service Engineer

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LX100/200 UNIT SPECIFICATIONS

- Operating Temperature: 15° - 30°C (59° - 86°F)
- Air / Sheath Pressure: 6-9 p.s.i. after stabilization
- High Voltage: 205 – 215v (on)
- Flow Rate: 10.2 – 11.4ml per 2 minutes
- Laser Split (green offset μ sec.): 25 – 50 μ s (micro seconds)
- Red (635nm) Laser Power Range: 7.5 – 9.5mW
- Green Uniphase (532nm) Laser Power Range: 10 – 12mW
- Green Coherent (532nm) Laser Power Range: 11.75 – 14.75mW
- DD, CL1, CL2 Temp: 18°– 35°C (Uniphase Configuration)
- DD, CL1, CL2 Temp: 15°– 35°C (Coherent Configuration)
- Board Temperature: 15 - 40°C
- DD A/D offset: 5 – 25 (laser off) / 5 – 150 (laser on)
- CL1, CL2 & RP1 A/D offsets: 5 – 25 (lasers off) / 5 – 35 (lasers on)
- DD: CV <16.0 / Gate Setting: 8,500 – 13,500 (Based on DD target of 10,000)
- CL1: CV <6.0 ** Mean +/- 4% (of target value)
- CL2: CV <7.5 ** Mean +/- 4% (of target value)
- RP1: CV <10.0 ** Mean +/- 4% (of target value)



LX100/200 6 & 12 Month PM List

The following items are replaced or cleaned during a 6 Month Preventive Maintenance Visit:

- | | |
|---|------------|
| 1. Syringe Teflon Cap | CN-0014-01 |
| 2. Rear Air Filter | CN-0001-01 |
| 3. Rear Filter (XY) | CN-0027-01 |
| 4. Analyzer Ventilation Filter (<u>Clean</u>) | CN-0002-01 |

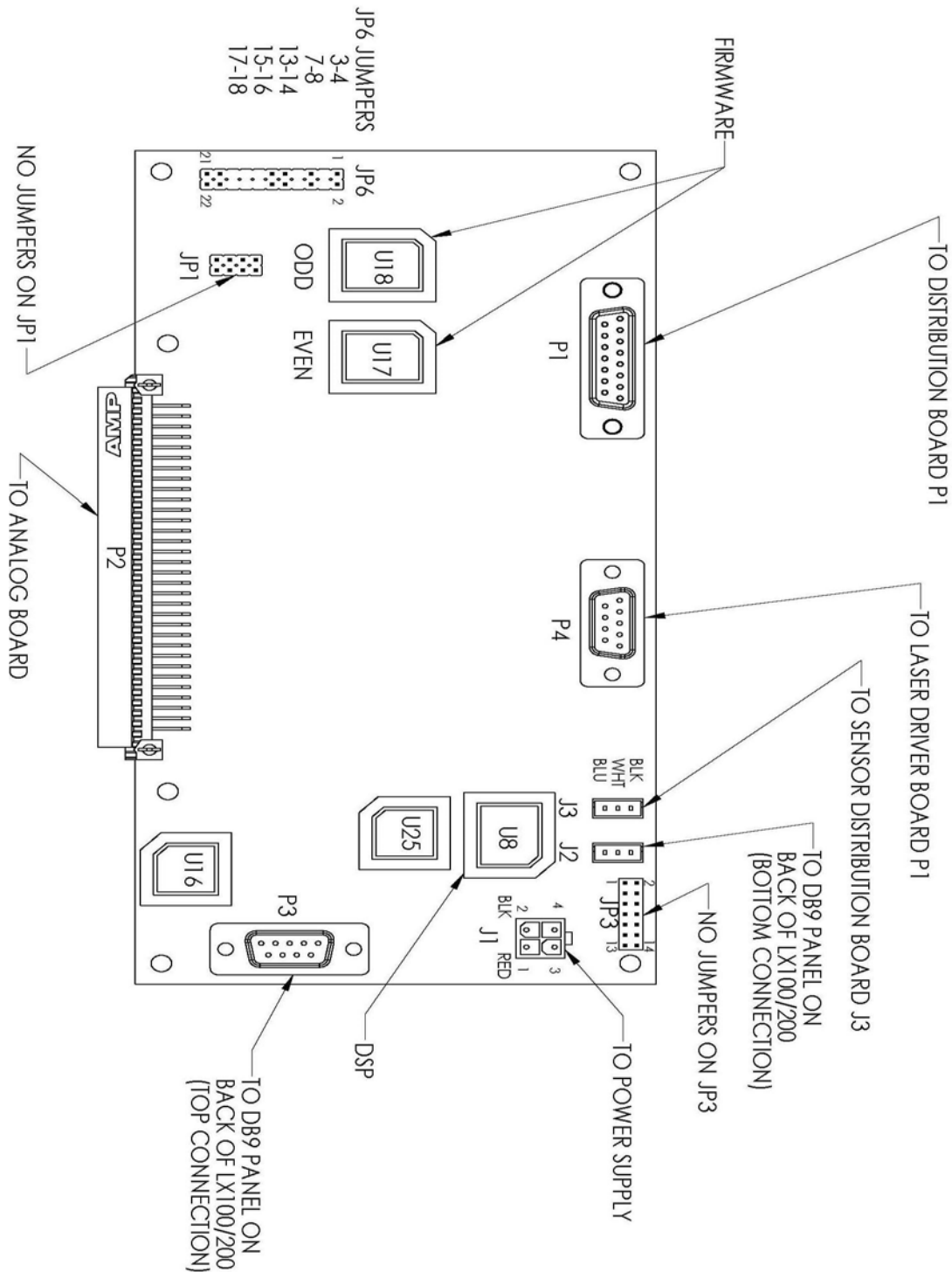
The following items are replaced or cleaned during a 12 Month Preventive Maintenance Visit:

- | | |
|--|-----------------------|
| 5. Orange Sheath Filter (Regular/Quick Disconnect) | CN-0028-01/CN-0010-01 |
| 6. Syringe Teflon Cap | CN-0014-01 |
| 7. Peek Tubing (10 inch) | FS-0096-01 |
| 8. Sample Loop Tubing (50 inch) | FS-0047-01 |
| 9. Sample Arm Tubing (5 inch) | FS-0048-01 |
| 10. Rear Filter | CN-0001-01 |
| 11. Rear Filter (XY) | CN-0027-01 |
| 12. Analyzer Ventilation Filter (<u>Clean</u>) | CN-0002-01 |

Following preventive maintenance, verification of unit functions and optical alignment should be performed.

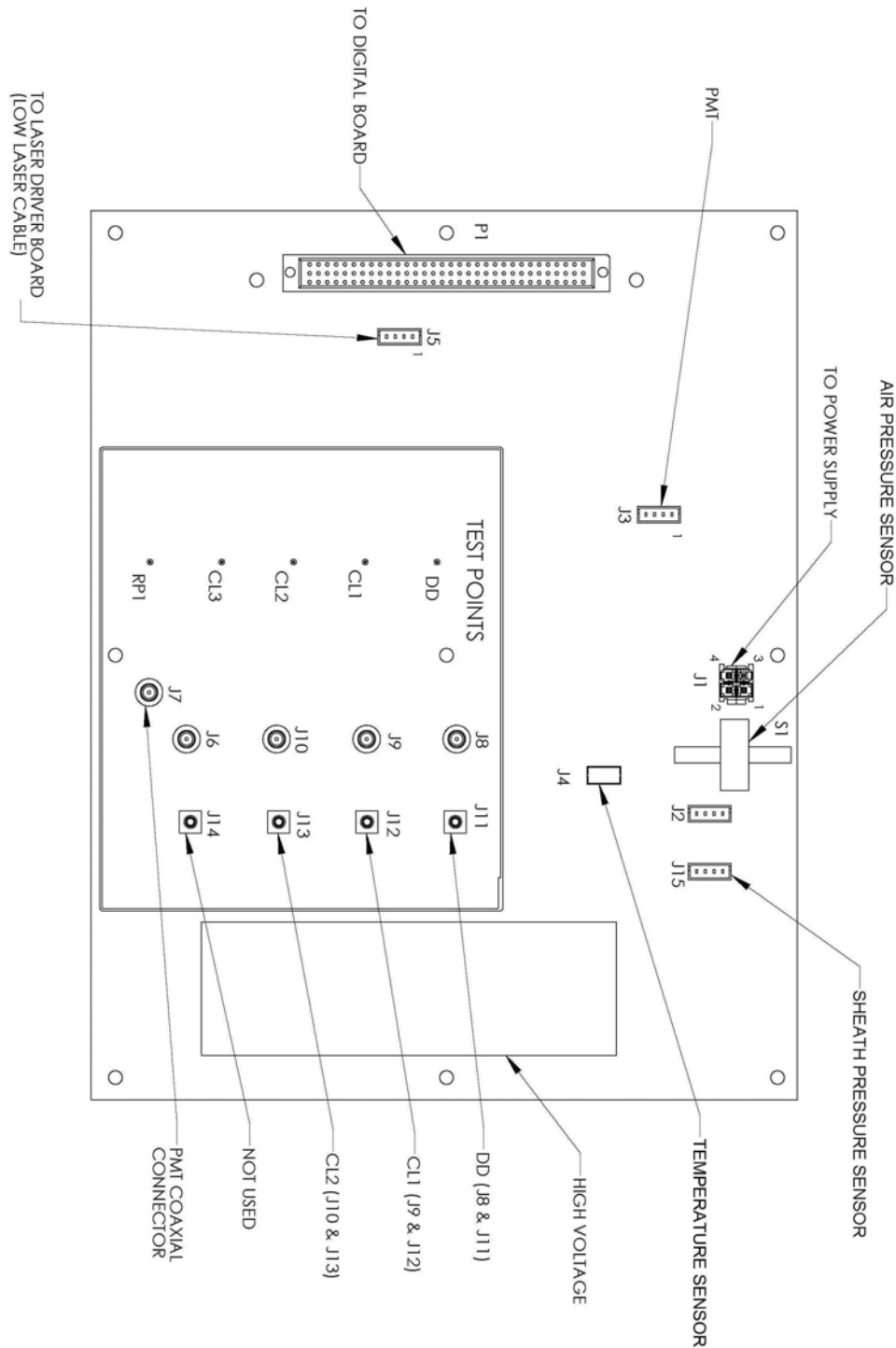
1) **DIGITAL BOARD:**

- Before replacing digital board, make sure that you can see the A.P.D. Breakdown and Gain values on the U-block. If not, write the values down; which can be found in the maintenance software. This is also where the new values will be entered.
- Upon replacing the digital board, enter & "apply or write" the new U-block A.P.D. gain (Vr) and breakdown (Vb) values, Serial #, Rev. & reset the A/D offsets ("Auto Offset").
- Verify that the Firmware version is correct for the customer's software application.
- For firmware versions 2.2.20 or higher, you will also have to set the three pressure parameters (manuf., calibr. & tolerance).
- Jumpers on JP6 from top to bottom: 3&4, 7&8, 13&14, 15&16, 17&18
- 5v input



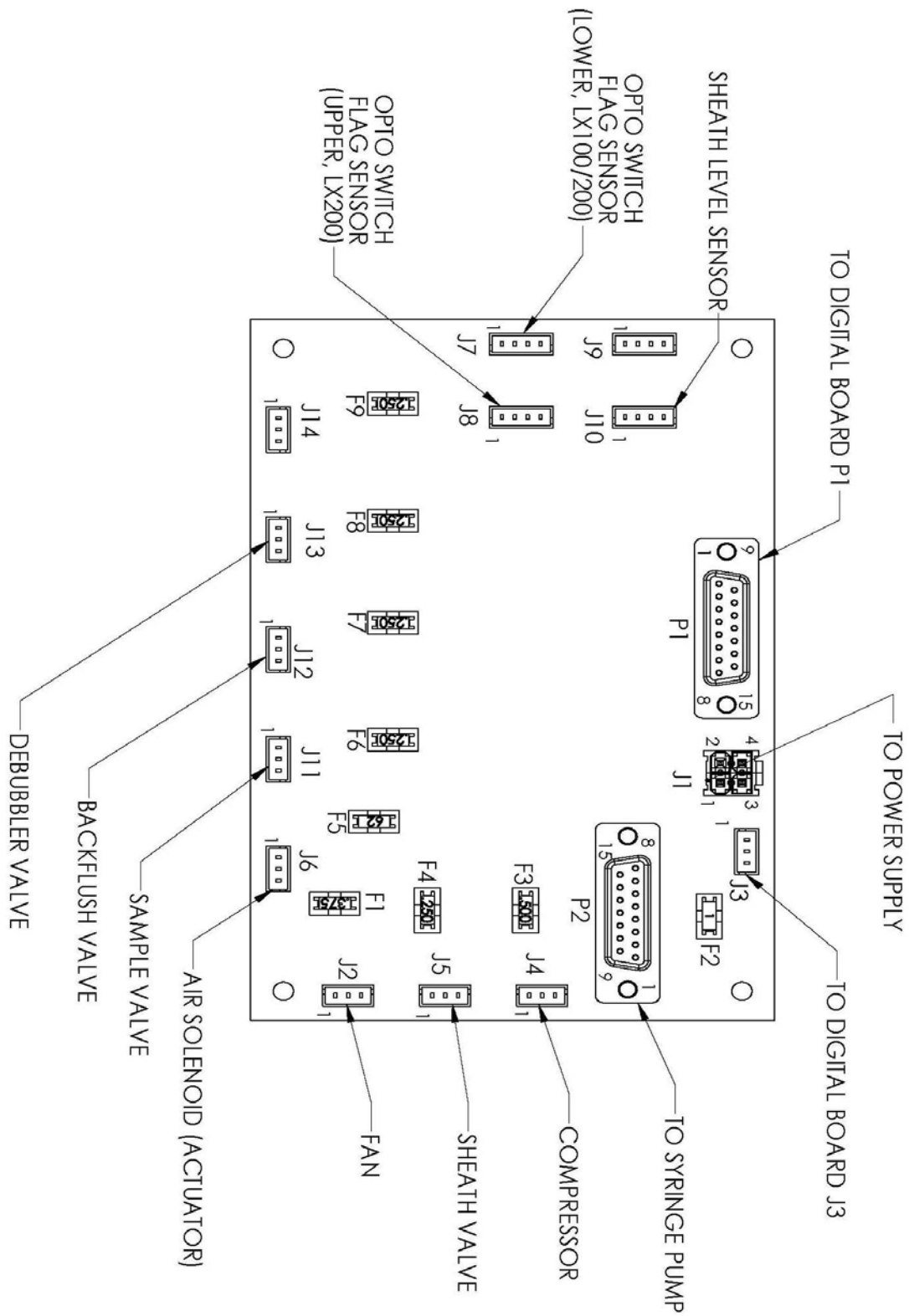
2) ANALOG BOARD:

- After replacing the analog board, reset the A/D offsets ("Auto Offset").
- Make sure to connect the black A.P.D. cables back in the right order (P1=J8&11, P2=J9&12, P3=J10&13. PMT=J7)
- Always make sure that you have not shorted out the two heat sinks when you replace the metal cover on the board. (If heat sinks are shorted out, your air pressure, A/D values and/or high voltage will not be correct)
- Be careful of the air pressure sensor on the board. The housing is not very hard to break; and once broken, the sensor will need to be replaced.
- Test Points For the O-Scope are located under the metal cover. They are directly adjacent to their respective A.P.D. cables (See diagram)
- +15 & -15v input

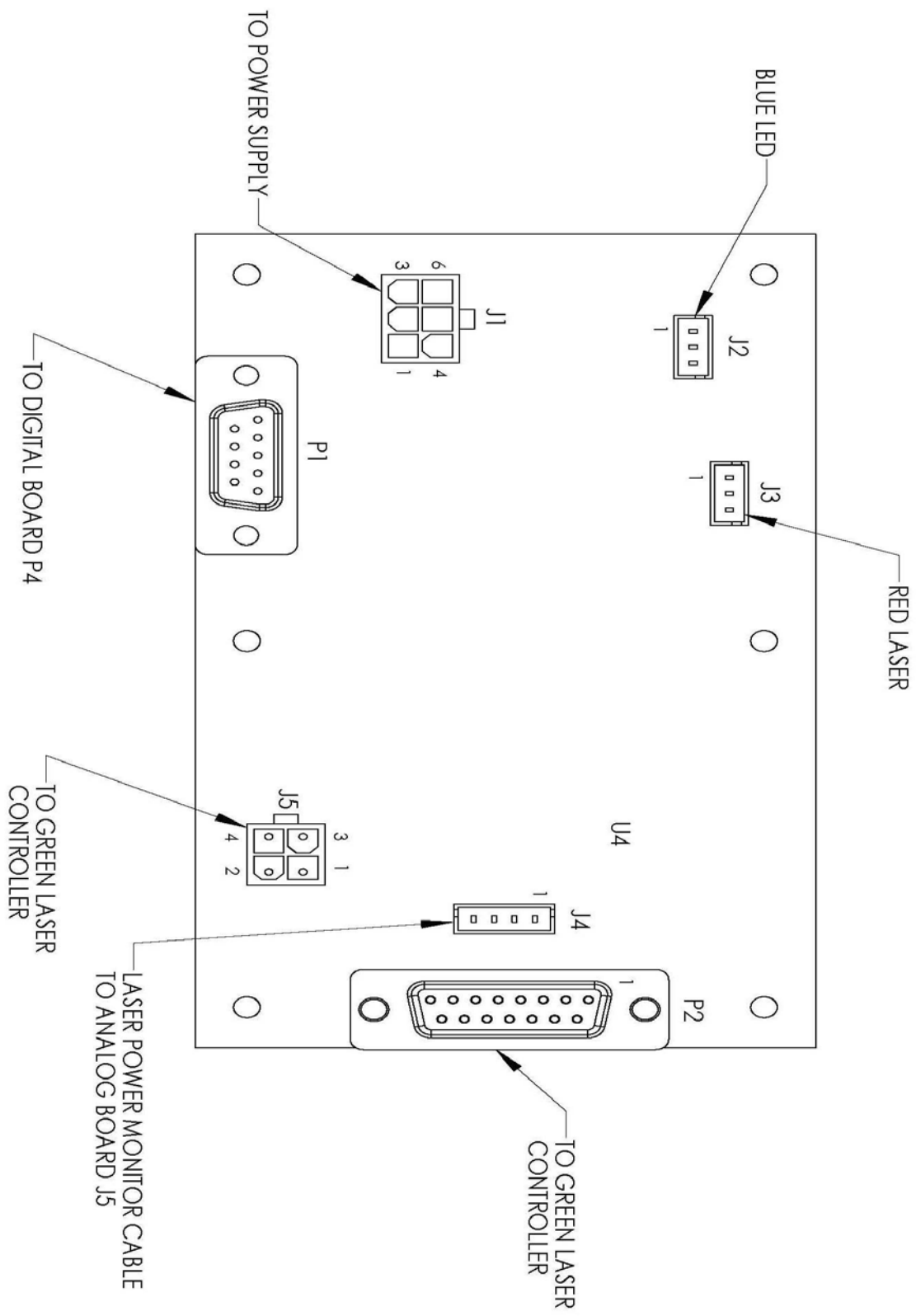


3) **DISTRIBUTION BOARD:**

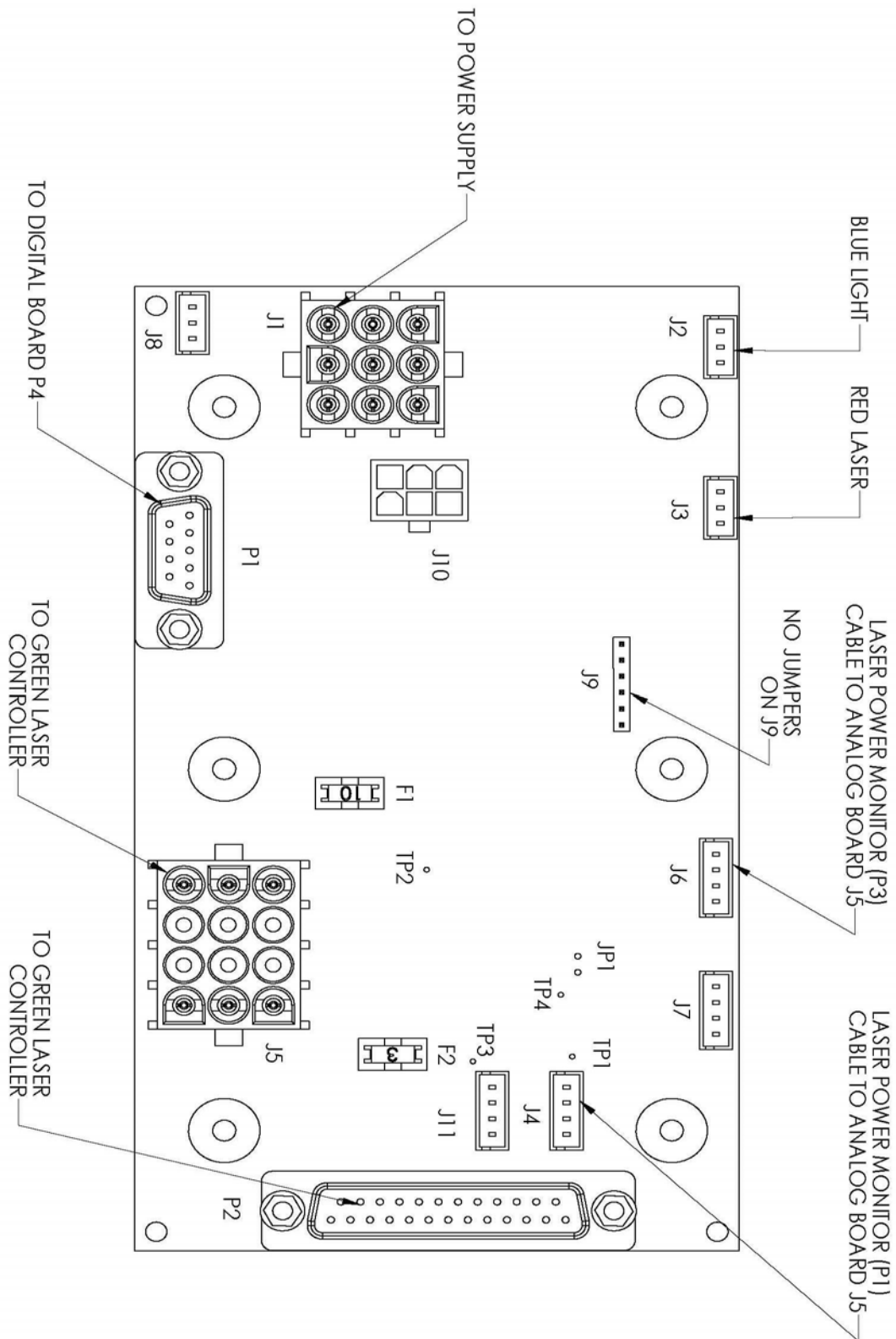
- Before replacing board, make sure to note connector locations
- Fuses & connectors are matched as follows(fuse/connectors): F1/J2, F2/J3, F3/J4, F4/J5, F5/J6, F6/J11, F7/J12, F8/J13, F9/J14
- 24v input



4) **LASER DRIVER BOARD (Uniphase Configuration):**
a) 5 & 15v input

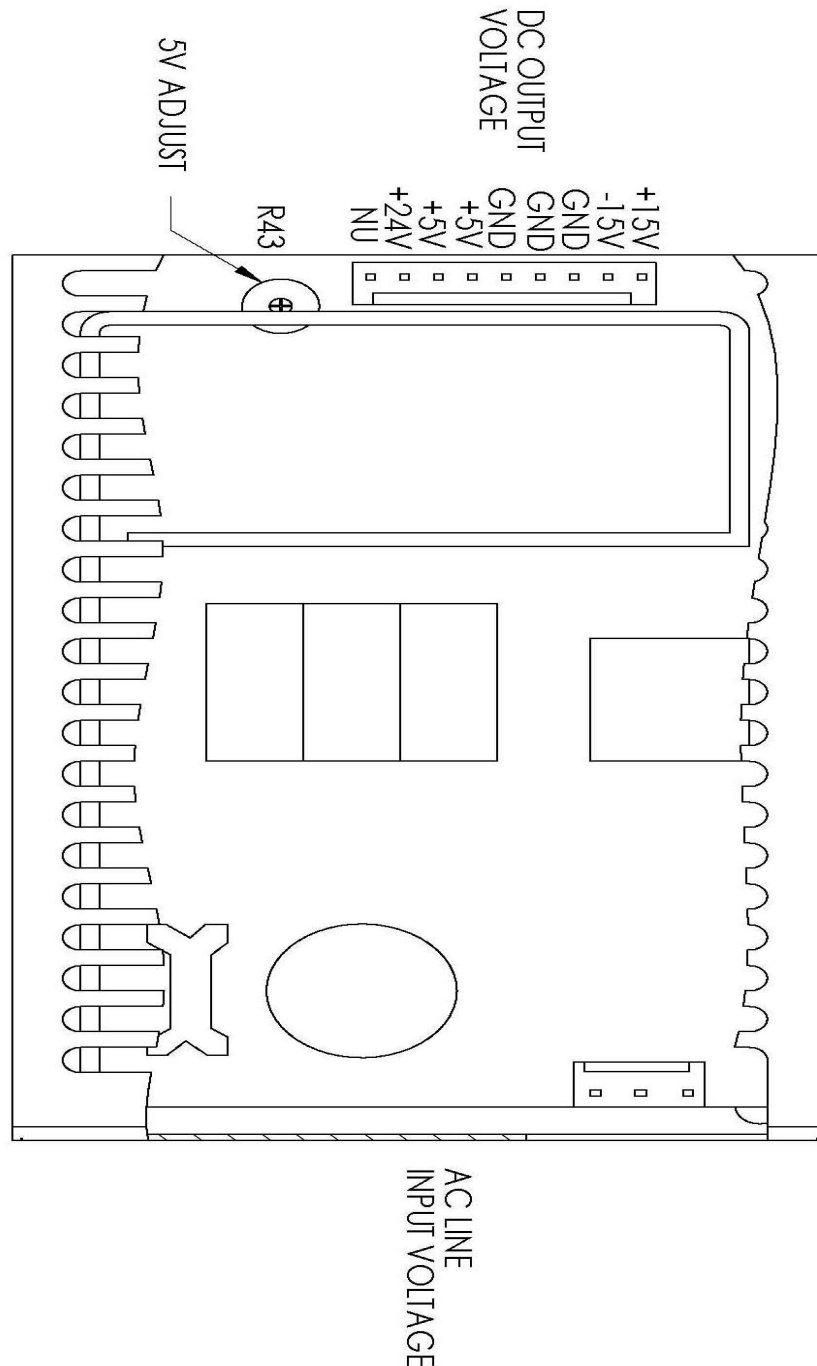


5) **LASER DRIVER BOARD / UNLIMITED (Uniphase & Coherent Configuration):**
a) 5 & 15v input



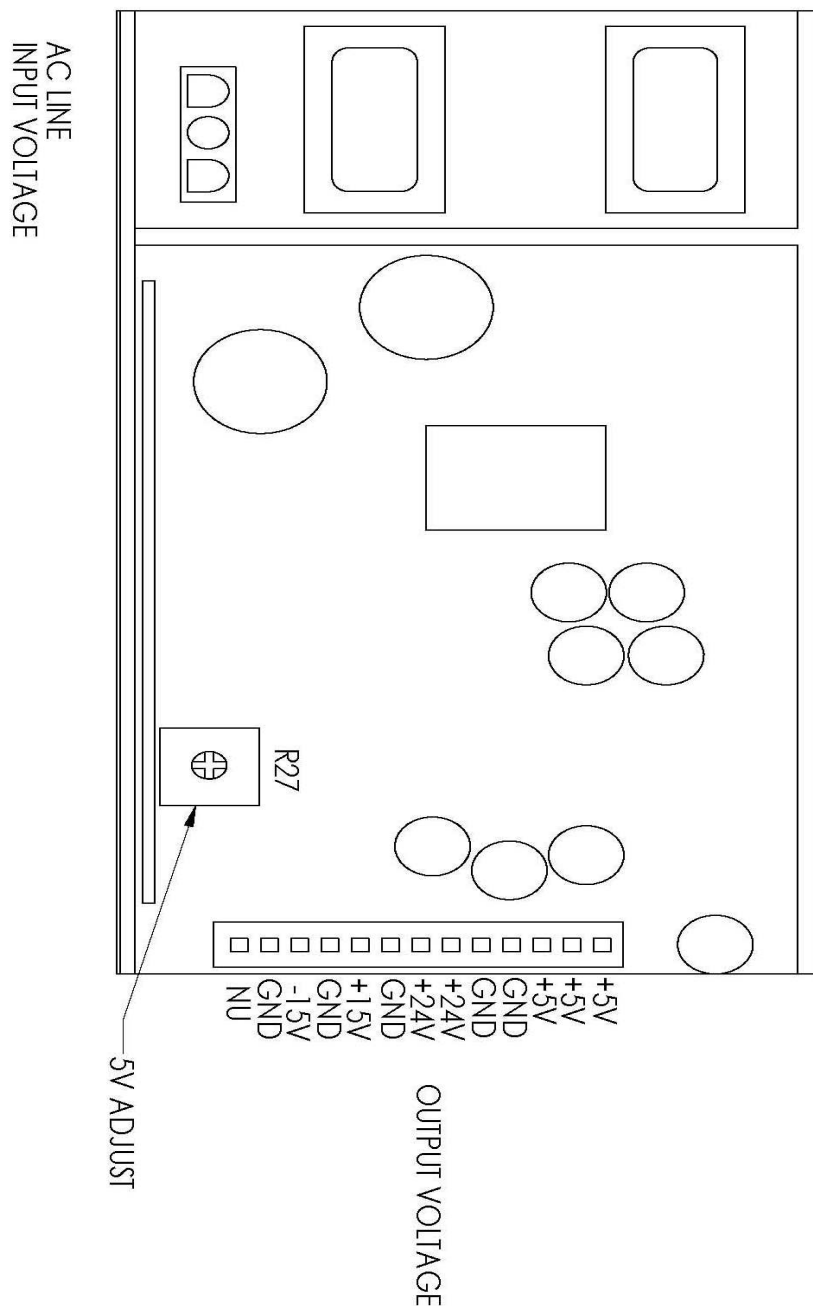
6) **POWER SUPPLY (Uniphase):**

- a) The 55 Watt power supply, which auto switches between 110 & 240v, is responsible for taking the units incoming line AC voltage, changing it to DC and distributing the appropriate individual voltages to their perspective boards. It's protected by 2, 2A 250v fast acting fuses.
- b) LX100/200 Power Supply Adjustment (Uniphase):
 - i) Turn off the instrument.
 - ii) Disconnect the green laser control box from the Laser Driver Board before performing this procedure.
 - iii) Turn the instrument back on and check power measurements.
 - iv) For systems using a Uniphase laser and a Condor power supply, the 5 Volt line of the power supply needs to be set between 5.50 and 5.55 Volts. This adjustment must be made before the laser controller is connected to the laser driver board. Place the voltage probes in the 9-pin connector of the power supply, making contact with the terminals for the red and black wires. Adjust pot R43 (near the 9-pin connector) on the supply until the voltage is in range (5.50 to 5.55 Volts)
 - v) Turn the instrument off and reconnect the control box cable previously removed



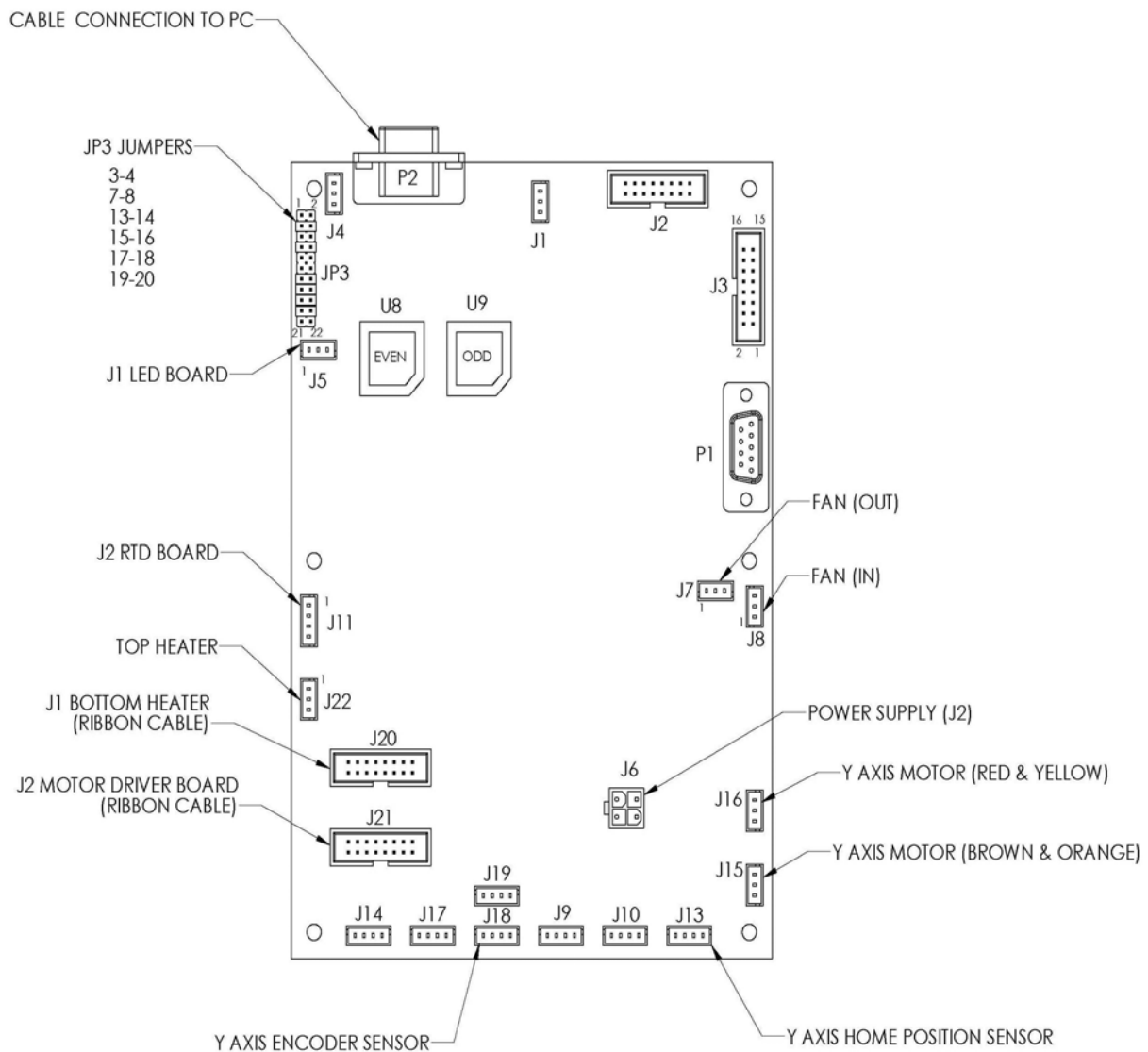
7) **POWER SUPPLY (Coherent):**

- a) The 100 watt power supply auto switches between 110 & 240v, and is responsible for taking the units incoming line AC voltage, changing it to DC and distributing the appropriate individual voltages to their perspective boards. It's protected by 2, 3A 250v fast acting fuses.
- b) LX100/200 Power Supply Adjustment (Coherent):
 - i) Turn off the instrument.
 - ii) Disconnect the green laser control box from the Laser Driver Board before performing this procedure.
 - iii) Turn the instrument back on and check power measurements.
 - iv) For systems using a Coherent laser and a Resonant Power Technology power supply, the 5 Volt line of the power supply needs to be set between 5.44 and 5.47 Volts. This adjustment must be made before the laser controller is connected to the laser driver board. Place the voltage probes in the 12-pin connector at the left of the power supply, making contact with the terminals for the red and black wires. Adjust the gray pot that is to the right of and above the 12-pin connector on the supply until the voltage is in range (5.44 to 5.47 Volts)
 - v) Turn the instrument off and reconnect the control box cable previously removed



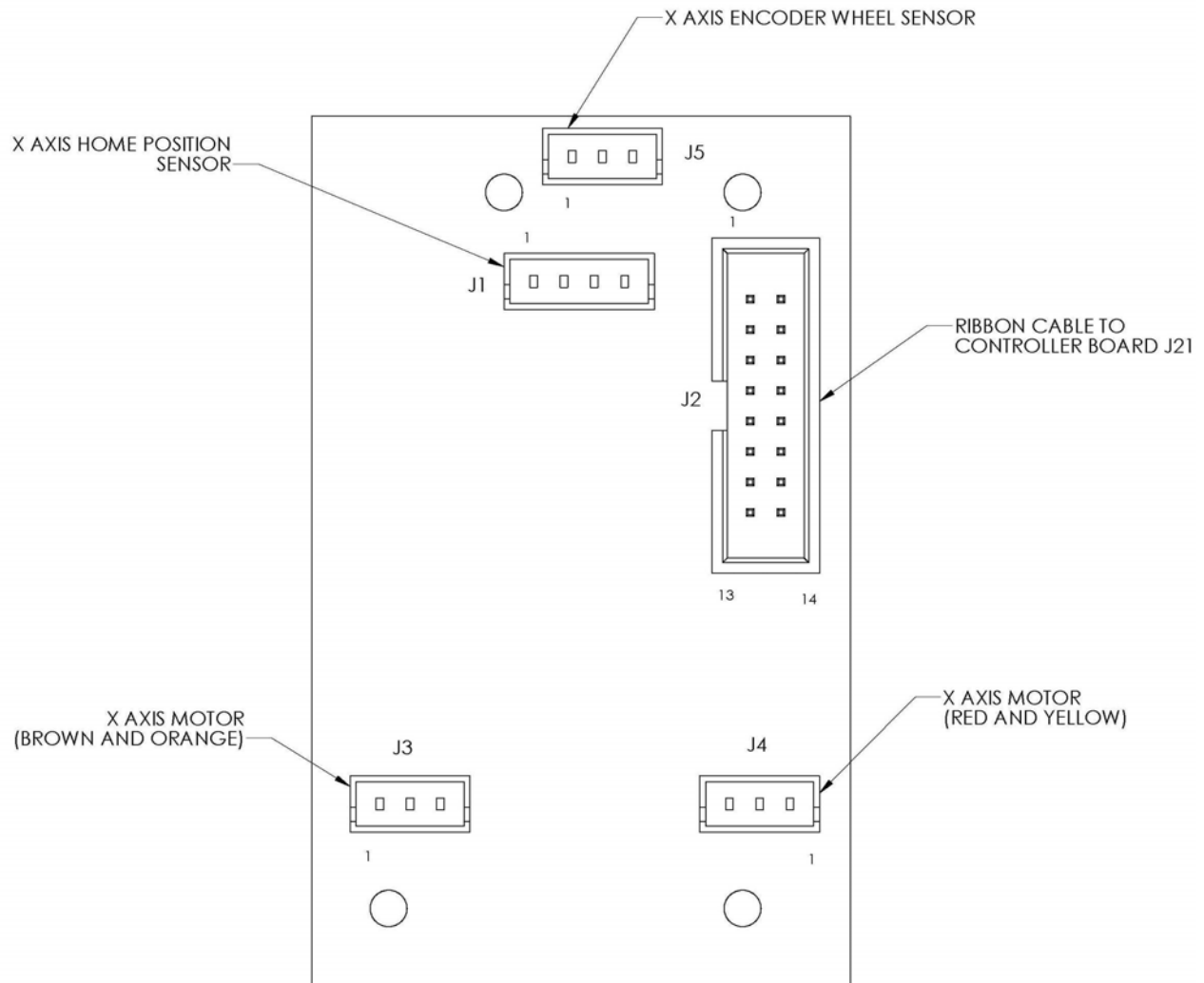
8) **XY CONTROLLER BOARD:**

- a) 24v input
- b) Houses NV Ram for storing unit information and system settings
- c) Firmware stored in U8 & U9



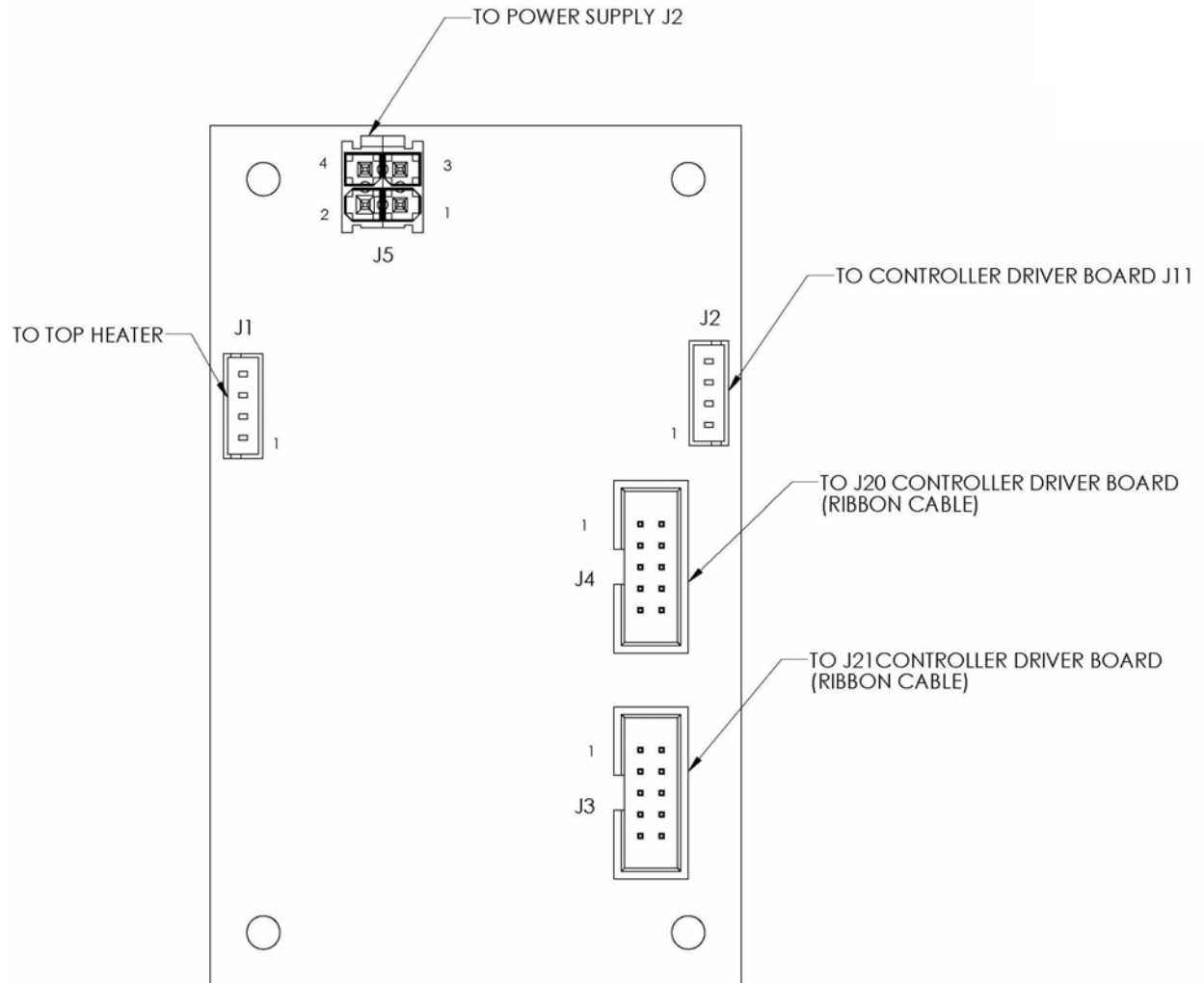
9) **XY MOTOR DRIVER BOARD:**

- a) 24v input
- b) Specifically for X-axis travel



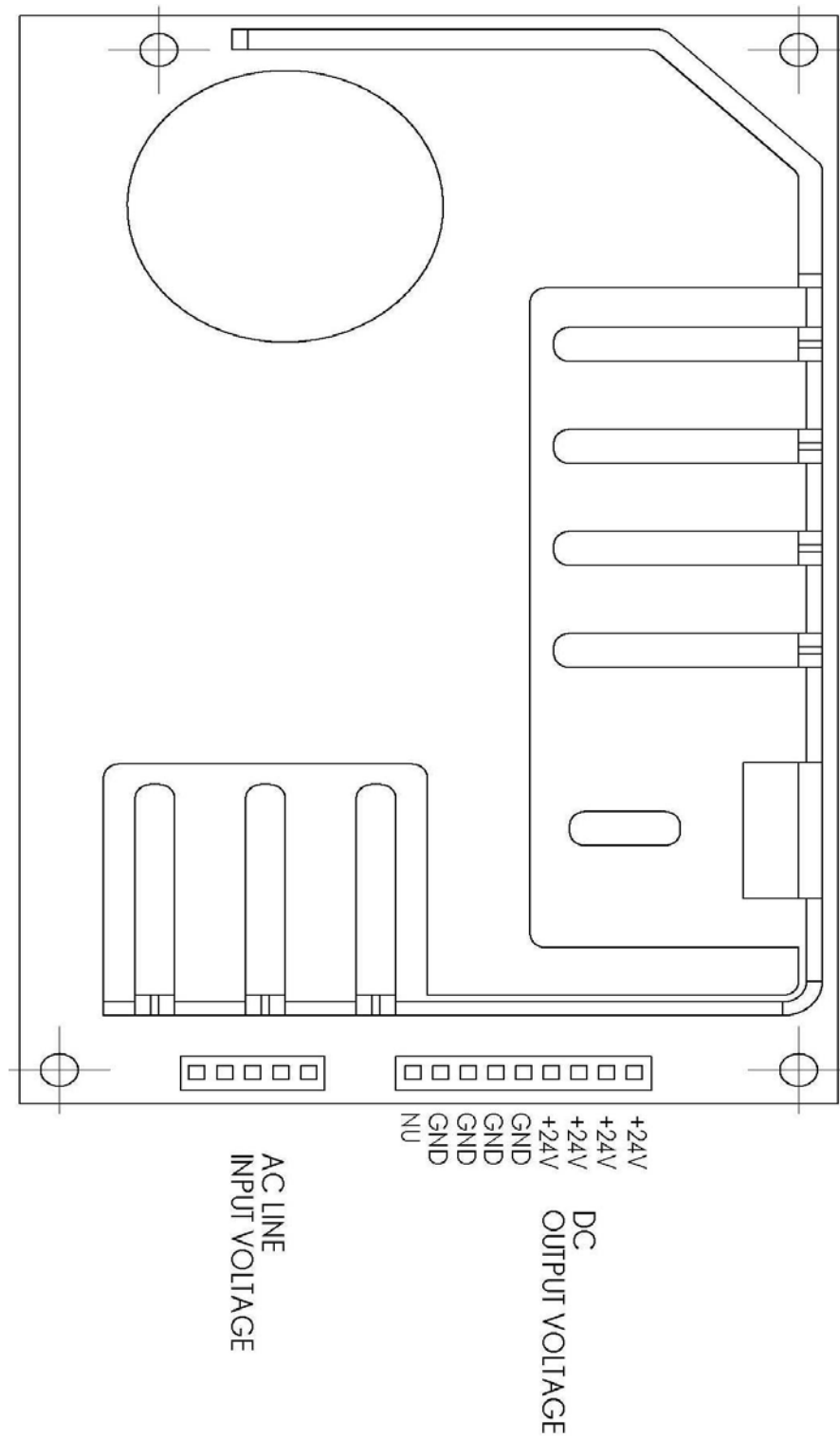
10) **XY RTD BOARD:**

- a) 24v input
- b) Takes heater information (analog in form), and converts it to a voltage for processing



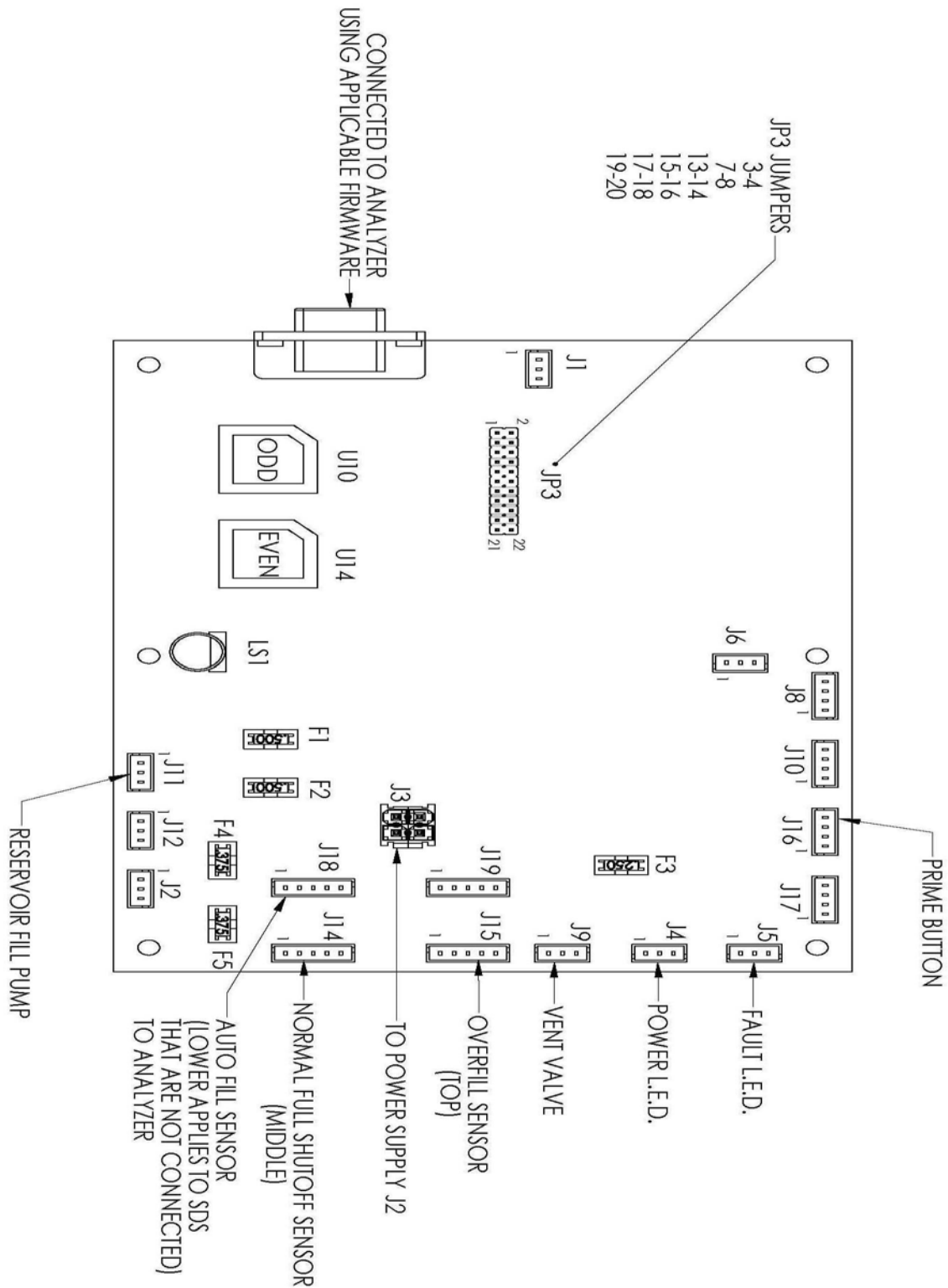
11) **XY POWER SUPPLY:**

- a) Auto switches from 100 to 240v
- b) Protected by two 3A 250v Fast acting fuses



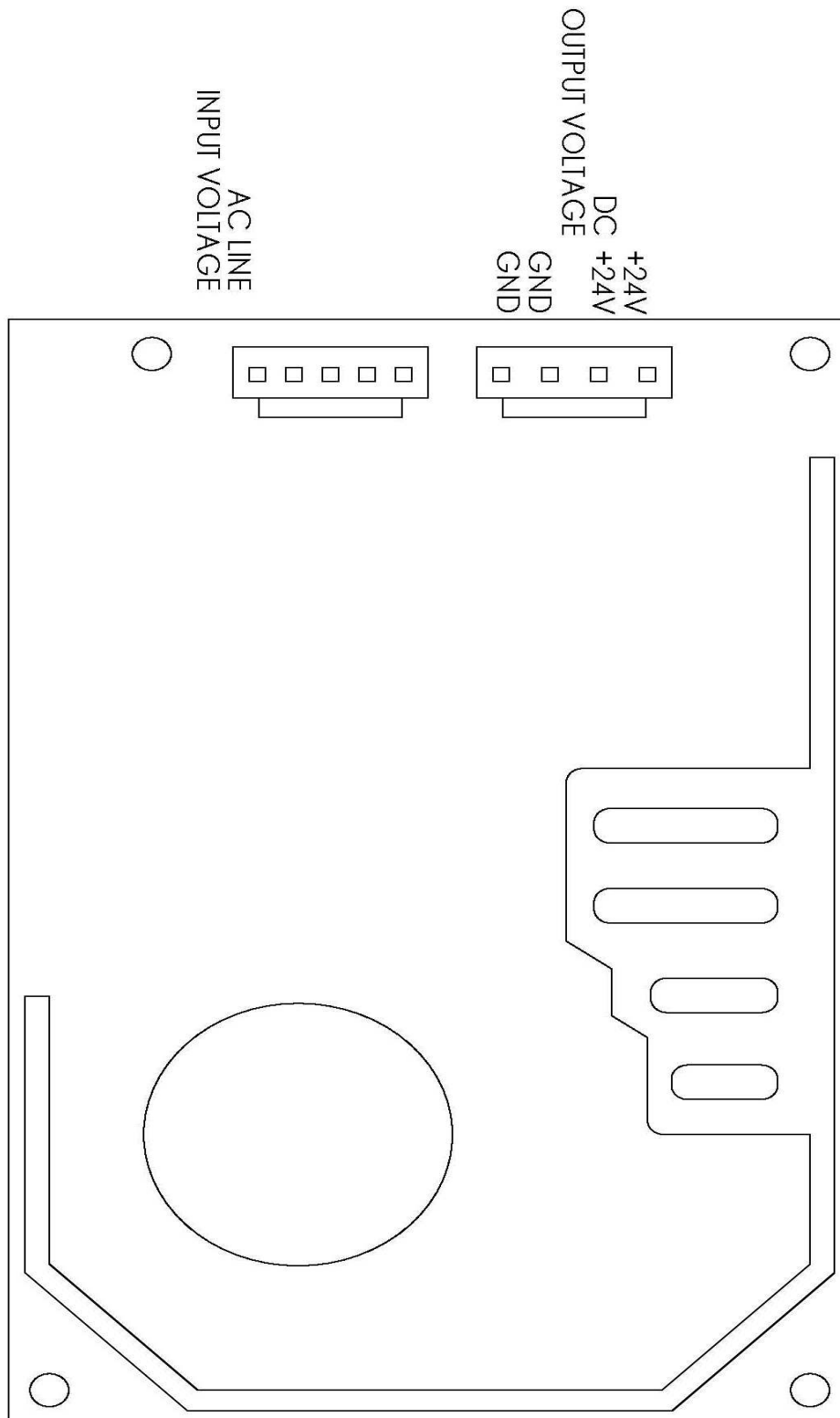
12) SD CONTROLLER BOARD:

- 24v input
- Houses NV Ram for storing unit information and system settings
- Firmware stored in U10 & U14
- Can be connected to an LX100/200 using the applicable hardware and firmware (SD and analyzer)

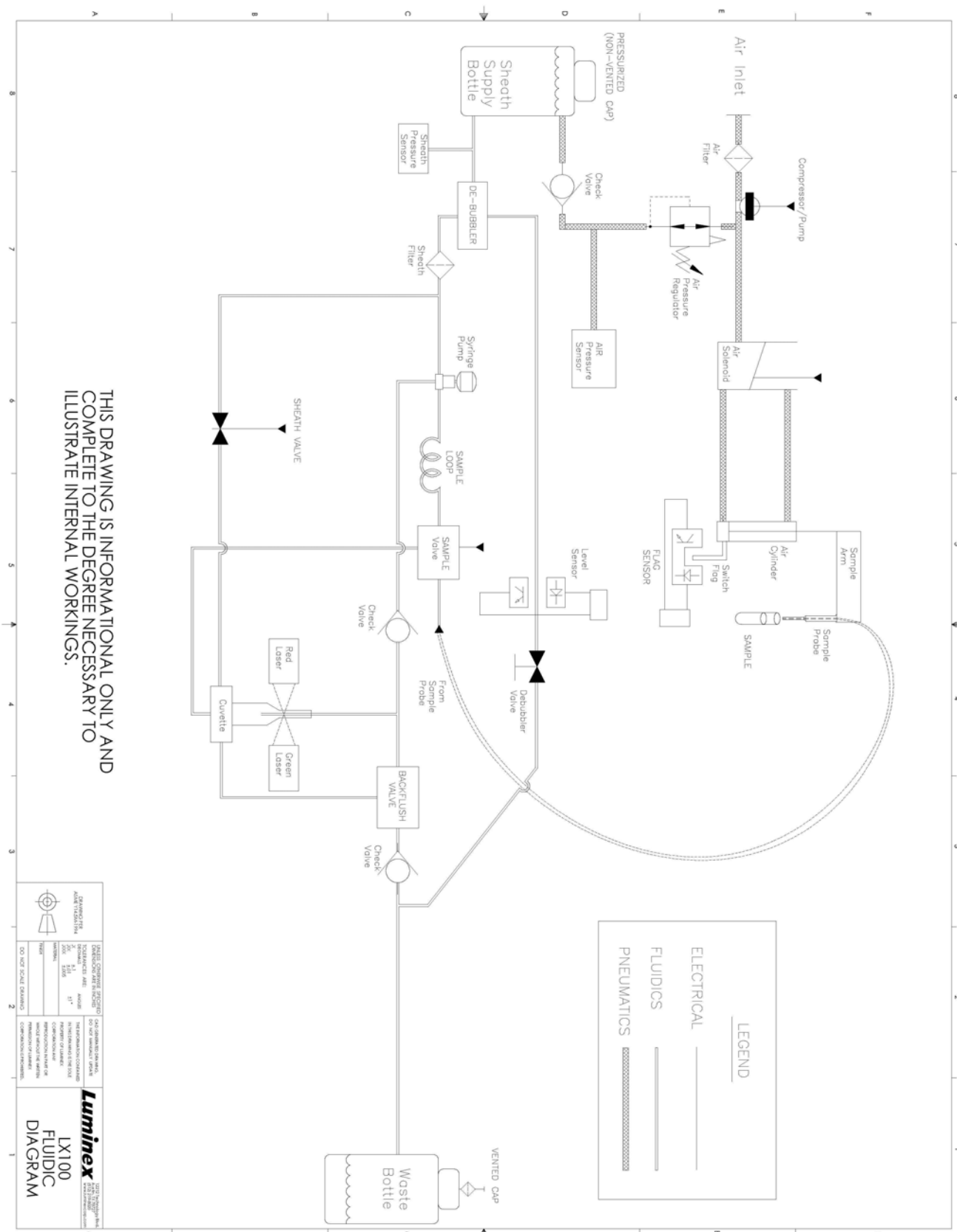


13) **SD POWER SUPPLY:**

- a) Auto switches from 100 to 240v
- b) Protected by two 2A 250v Slo-blo fuses



14) LX100/200 FLUIDIC DIAGRAM:



Luminex Lx100 Fluidics

Power Up

1. Turns on compressor to pressurize sheath & air
2. Opens sheath valve to start sheath flow
3. Takes full stroke from sheath -> syringe -> sample loop -> cuvette
4. Opens backflush valve for ~2.0 seconds and allows sheath to flow through
5. Closes backflush valve so sheath flows through top of cuvette again
6. After ~5 seconds, closes sheath valve (stops sheath flow)
7. Waits ~50 seconds for electronics to stabilize before turning on lasers
8. Turns lasers on

Prime

1. Turns on compressor to pressurize sheath & air
2. Opens sheath valve to start sheath flow
3. Allows sheath to flow through cuvette ~45 seconds
4. Takes full stroke from sheath -> syringe -> sample loop -> cuvette twice (removes bubbles from syringe & bubble trap)
5. Closes sheath valve (stops sheath flow)

Wash

1. Turns on compressor to pressurize sheath & air
2. Opens sheath valve to start sheath flow
3. Lowers sample probe
4. Takes 200uL from sheath -> syringe
5. Takes 200uL from sample -> sample loop
6. Raises sample probe
7. Takes sheath + wash from syringe + sample loop -> cuvette
8. Quickly alternates running sheath through top and bottom of cuvette 5 times; finishes by running sheath through top of cuvette
After ~5 seconds, closes sheath valve to stop sheath flow

Backflush

1. Opens sheath valve
2. Turns on compressor to drive the sample probe
3. Takes full stroke from sheath -> through top of syringe -> cuvette -> through backflush port twice
4. Closes sheath valve

WarmUp

1. Turns on compressor to pressurize sheath & air
2. Checks if lasers are on or off
3. If lasers are off, turns lasers on and remains in Warmup state for set warmup time
4. If lasers are already on, exits warmup

Luminex Lx100 Fluidics

Start

1. Turns on compressor to pressurize sheath & air
 2. Checks if lasers are on or off
 3. If lasers are off, turns lasers on and remains in Warmup state for set warmup time
 4. If lasers are already on, exits warmup
 5. Opens sheath valve to start sheath flow
 6. Waits for pressure to stabilize
 7. Lowers sample probe
 8. While sample probe is coming down, takes 50uL from sheath -> syringe, which will be injected into the cuvette following the sample
 9. Sample agitation - sucks up, then immediately expels 15uL of sample
 10. Draws the requested amount of sample into sample loop
 11. Raises sample probe
 12. Quickly preinjects the sample up to the needle within the cuvette
 13. Slows the inject rate & lets the sample stream stabilize
 14. Begins capturing bead events
 15. When the syringe has been fully depressed, turns off bead capture
- After ~10 seconds, closes sheath valve**

Cancel

1. Turns off bead capture
2. Lowers sample probe
3. Quickly injects any remaining sample into the cuvette
4. Takes ~300uL from sheath -> syringe
5. Runs ~217uL from syringe -> sample loop -> cuvette
6. Runs remaining sheath from syringe -> sample loop -> sample probe
7. Raises sample probe
8. After ~10 seconds, closes sheath valve (stop sheath flow)

Soak

1. Turns on compressor to pressurize sheath & air
2. Opens sheath valve to start sheath flow
3. Lowers sample probe
4. Takes 250uL from sample -> sample loop
5. Raises sample probe
6. Runs sample from sample loop -> cuvette
7. Quickly alternates running sheath through top and bottom of cuvette 5 times; finishes by running sheath through top of cuvette
8. After ~5 seconds, closes sheath valve to stop sheath flow

Luminex Lx100 Fluidics

Sanitize & Alcohol Flush

1. Closes sheath valve (stops sheath flow)
2. Turns on compressor to drive the sample probe
3. Lowers sample probe
4. Takes ~200uL of sample from sample probe -> sample loop -> cuvette 4 times (resident fluid in the cuvette will be pushed through the top of the cuvette)
5. Takes ~200uL of sample from sample probe -> sample loop
6. Raises sample probe
7. Pauses for 2 minutes
8. Opens sheath valve to start sheath flow
9. Runs the 200uL sample from sample loop -> cuvette
10. Takes full stroke from sheath -> syringe -> sample loop -> cuvette 10 times
11. Lowers sample probe
12. Takes full stroke from sheath -> syringe -> sample loop -> sample probe 3 times
13. Raises sample probe
14. After ~10 seconds, closes sheath valve (stop sheath flow)

Drain

1. Turns on compressor to drive the sample probe
2. Take ~200uL air from sample probe -> sample loop -> cuvette 11 times (fluid in the cuvette will be pushed through the backflush port)
3. Lowers sample probe
4. Takes full stroke from sheath -> syringe
5. Runs ~3/4 of sheath through syringe -> cuvette -> backflush port
6. Runs other 1/4 of sheath through sample loop -> sample port
7. Closes the backflush valve
8. Opens sheath valve to start sheath flow through top of cuvette
9. After ~4 seconds, closes sheath valve to stop sheath flow
10. Raises sample probe
11. Closes sheath valve

15) GREEN LASER CONTROLLER (Uniphase):

- a) When the laser is on and has stabilized, only three of the controller lights should be on, and steady. The two on the far left and the one on the right (See below). After stabilizing; if either red light comes on, or if any light is blinking, laser replacement is almost certain if the 5v input to the controller is correct
- b) The function of the lights are as follows (Left to Right):

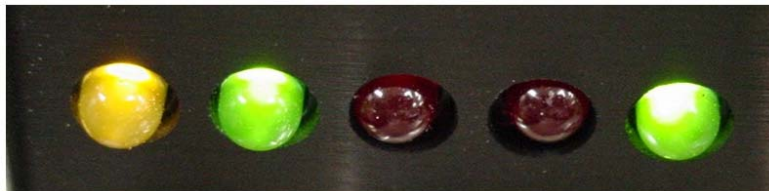
Yellow—Emission, illuminated 5 seconds before the laser emission. Remains on while the laser emission is ON

Green-- Laser in spec. Illuminated when the laser power is within +/- 5% of the set point

Red--Thermal fault, illuminated when either of the TE control loops cannot maintain its temperature set point

Red-- Internal fault, illuminated when any fault is encountered (i.e. thermal fault, laser out of spec.)

Green-- Unit power, illuminated whenever input DC voltage is supplied to the controller

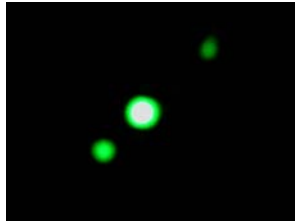


16) PRESSURE SETTINGS (Firmware Version 2.2.24 & Higher):

- a) This firmware requires that the current sheath pressure is within 5% of the calibrated sheath pressure in order to allow a user to perform various nominal functions (wash, sample run, prime command, etc.) If it's out of the 5% window, the function will time out and an error will occur as the unit is not able to achieve the required pressure value. There are few ways to adjust these settings, the two most common are listed below
 - i) IS software has 4 property values that allow one to view and adjust the pressure settings. These property values are 1267, 1268, 1269 & 1002. 1267=Manufacturing P.S.I., 1268=Last Calibration P.S.I., 1269=% Tolerance (5%) & 1002=Current Sheath Pressure. The best way to start off is to set both 1267 & 1268 the same as the current sheath pressure value in 1002, which is read with the compressor on and after the pressure stabilizes. From there, once you calibrate, the settings will automatically update. ****You must have the "Advanced" tab available to you in the software in order to perform the adjustments.**
Example: Type the property location, 1267 for example, into the "Property ID" box and select "Get". Your information will show up in the "Property Value" box. If the number is correct, leave it alone; if not, change to the correct value then select "Set" (repeat steps for other property values)
 - ii) A second is to use "LXR Maintenance" and make the changes under the "Manufacturing Settings" section. Simply "Double click" on the appropriate box, change the value and select "Write" (make sure to take your current sheath pressure value with the compressor on and the pressure stabilized)

17) GREEN LASER (Uniphase):

- a) Good laser pattern (Shown)
- b) The diameter of the laser beam (seen on the PMT while running Laser Alignment Solution), can be adjusted using the “Vertical” screw of the 4mm lens of the Reporter channel
- c) 5v input



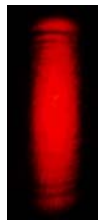
18) GREEN LASER (Coherent):

- a) Good laser pattern (Shown)
- b) The diameter of the laser beam (seen on the PMT while running Laser Alignment Solution), can be adjusted using the “Vertical” screw of the 4mm lens of the Reporter channel
- c) 5v input



19) RED LASER:

- a) Good beam pattern (Shown)
- b) For proper operation, beam pattern must be vertical when installed
- c) Has potentiometer with which minor power adjustments can be made
- d) Adjusting the potentiometer may help to correct a shifting of the entire bead map (“**Caution**, only make very minute adjustments. Recalibrate the classification channel, and verify that the power output is within the specified tolerance)
- e) 5v input



20) SYRINGE PUMP:

- a) After replacing, always perform a “reset” to ensure proper operation. This can be found in the “Syringe Operations” section of LXR Maintenance. (NOTE: The unit “**Must**” be powered down and up again for the setting to take effect.)

21) CUVETTE INJECTION NEEDLE:

- a) This can get clogged on occasion. Verify that the “Needle” is clear by removing the peek tubing w/fitting, from the underside of platform, and run a wash to verify that a fine stream comes out of the “hockey puck”, where the tubing was removed, for most of the cycle as well as steady drops or spray coming from the end of the peek tubing while the syringe is on its way up (be sure to place a paper towel under the platform to catch the fluid).
(NOTE: This will “NOT” tell you whether the tip of the “Cuvette” itself is clogged)

22) BUBBLE TRAP LEVEL SENSOR:

- a) There are two settings on this switch. The upper switch is to the right and the lower switch is to the left.

23) U-BLOCK (A.P.D. Boards & Flex. Temp. Circuit):

- a) The U-Block contains 3 A.P.D. boards and each has its own “Breakdown & Gain” voltage. If any of the photo diodes on the boards or the U-Block is replaced, you must enter the new values into the software via “LXR Maintenance”. You must then reset the A/D offsets. (Troubleshooting Hint: to verify that the voltage is getting to the individual APD board, run a sample and check the voltage on the center leg of the “Push On” style connector, on the A.P.D. board. It should equal the current “Gain” value in the optical detector settings (Not the gain value of the U-block itself).
- b) A.P.D. cable connections are as follows: (P4=DD, P5=CL1 & P6=CL2)
- c) The Temp. Flex circuit reads the individual temperatures for DD, CL1 & CL2 and is connected to the Analog board on J4.

TROUBLESHOOTING TIPS

***Always begin by checking the basics, which can save you, time in troubleshooting. These include, but are not limited to: PC/Unit set up and connections, Flow rate, Laser power, A/D offsets, Probe height and the information displayed in the software program (Temperatures, Events, Pressures, etc.) **Suggestions are in no particular order of importance.**

There are 5 “Start Up” indicators to check for when powering up the LX100 or 200

- 1. Blue light (immediately comes on)**
- 2. Fan (immediately comes on)**
- 3. Syringe pump (after a small delay takes 1 full stroke)**
- 4. Compressor (comes on when syringe takes the stroke)**
- 5. Drip in waste bottle (unit does a prime upon start up...syringe stroke)**

24) Analyzer connection errors:

- a) Make sure connections in the back of the unit or pc are completely seated
- b) For 1.7, verify the software com port settings are correct under “Analyzer, Set Up”
- c) Run Configuration Wizard when connecting with IS 2.1 & 2.2 software
- d) USB driver not installed or needs to be reinstalled. (if applicable)
- e) Digital board
- f) Firmware is incompatible to the software you are attempting to connect to
- g) Firmware is incompatible with the type of communication cable you are attempting to connect with
- h) Serial number is missing from NV Ram on the Digital board, or it isn't of correct format
- i) Whenever you upgrade the SD firmware to 2.0.5, you will need to restart the SD; otherwise you may run into some new problems such as the analyzer not starting up properly or no connectivity to the software

25) Air and/or sheath pressure reads zero or is incorrect:

- a) Sheath bottle not plugged in or cap is loose
- b) Check that the metal cover on the analog board isn't shorting out the heat sinks next to it
- c) SD or sheath bottle may have a broken fitting inside (plastic fitting). Unplug SD or bottle from analyzer and see if air pressure rises
- d) Damaged or missing o-ring on green quick disconnect from the bottle or SD. Unplug SD or bottle from analyzer and see if air pressure rises
- e) Compressor or fuse
- f) Air pressure regulator
- g) Analog board
- h) Yellow restrictor between compressor and air pressure regulator is clogged
- i) Verify syringe pump takes full stroke during start up (which allows compressor to turn on)

26) Unit has no fluid going to waste during wash:

- a) Backflush valve or fuse
- b) Sheath valve or fuse
- c) Waste bottle not plugged in
- d) Sever clog at the top of cuvette

27) Unit won't calibrate after replacing digital board:

- a) Make sure that you have re-entered the correct U-block A.P.D. "breakdown and gain" values in the software
- b) Make sure you have re-set the A/D offsets after entering new "breakdown and gain" values (high offsets make it difficult to calibrate)

28) Sample arm goes down but won't come up (Software indicates "busy" or "pressurizing" and won't continue:

- a) Flag sensor may need adjusting (sometimes you can tell this if you wiggle the needle linkage while the arm is down and see if it continues)
- b) Bad/leaking air solenoid or fuse (check the vent hole(s) to see if they are leaking when the arm is in the up or down position)
- c) Incorrect needle height, adjusted too low (The sample arm flag on the air cylinder has not gone down far enough to reach the flag sensor, which tells the unit that the arm is down and allows it to continue (Not applicable when performing a "Wash"). Make sure that when the sample arm is going down, the needle and/or sample arm isn't hitting something to make it stop prematurely)

29) No events / low events:

- a) Bad red laser (weak or dead)
- b) End of waste line is submerged causing back pressure
- c) Red laser not on
- d) Sheath, backflush valve or fuse (Check for drip going to waste during prime or while running sample)
- e) Waste line not plugged in or seated completely
- f) Waste bottle full or cap not vented
- g) Cuvette and/or cuvette needle clogged
- h) Sample intake needle clogged or partially clogged (verify that the unit takes up the sample)
- i) Needle height too low (needle is hitting the bottom of sample holder or plate in XY)
- j) DD A.P.D. "Gain" value is too low to see events (This can sometimes happen after replacing the analog board. Set the gain value in the optical detector settings to about 90 after starting sample or calibration)
- k) Sample valve or fuse
- l) No high voltage or high voltage incorrect

****Several of these possibilities display the symptom of air being drawn into the syringe during use**

30) Classification and/or Reporter A/D offsets won't reset or fluctuate too much:

- a) Analog board
- b) Laser (beam pattern may be irregular causing stray light)
- c) U-block or individual A.P.D. board
- d) A.P.D. cables
- e) PMT
- f) Digital board
- g) Dirty lenses or cuvette

31) No change to the signal on the scope when changing an A.P.D. Gain voltage:

- a) Analog board
- b) A.P.D. cable(s)
- c) A.P.D. board. Replace the U-block or the individual board that won't change. *If an A.P.D. board is replaced, remember to enter the new "Gain" and "Breakdown" values of the board into the software and mark the change on the U-block
**To help determine if it's the Analog board or U-block, turn off the power to the instrument and switch the problem channels cables, on the U-block, with another channel (Do not switch them on the analog board end). If the problem still exists on the same channel, it points more to the Analog board or cable; if not, it's possibly a problem with the individual channel's A.P.D. board or diode
- d) Verify that the O-scope/meter isn't on "Hold"

32) Dripping from sample needle or complaint of different levels of "Spitback" volume in plate:

- a) 3-Way sample valve or fuse
- b) Severe clog in cuvette
- c) Partially blocked sample probe

33) Nothing going to waste:

- a) Backflush valve or fuse
- b) Waste bottle not plugged in
- c) Sheath Valve or fuse
- d) Severe clog in cuvette

34) Excessive fluid going to waste or drip in waste container when unit is idle:

- a) Debubbler valve or fuse
- b) Sheath level sensor on B.B. trap (First check to see if the small red l.e.d. is lit on the sensor. Although not obvious, it can be seen when viewed from above. The sensor may need to be repositioned on the neck of the bubbletrap top. Try moving it around and see if it fixes the problem)
- c) Backflush valve or fuse
- d) Sheath valve or fuse

35) Unit doesn't get beyond pressure when attempting to run assay, wash etc:

- a) The unit's pressure settings may have to be re-done (refer to #16 for adjusting pressure settings)
- b) Compressor or fuse
- c) Sheath bottle not plugged in or cap loose
- a) The system may not be within the 5% tolerance range of the calibrated sheath pressure
- b) Air solenoid or fuse
- j) SD or sheath bottle may have a broken fitting inside (plastic fitting). Unplug SD or bottle from analyzer and see if air pressure rises
- k) Damaged or missing o-ring on green quick disconnect from the bottle or SD. Unplug SD or bottle from analyzer and see if air pressure rises
- l) Air pressure regulator
- m) Yellow restrictor between compressor and air pressure regulator is clogged

36) Unit displays “sheath empty” in software:

- a) Sheath bottle / SD empty
- b) Sheath level sensor (First check to see if the small red l.e.d. is showing on the sensor. Although not obvious, it can be seen when viewed from above. The sensor may be positioned too high on the neck of the bubbletrap top. Try to lower it down and see if it fixes the problem)

37) Seeing or pulling air into syringe:

- a) Needle height incorrect (needle adjusted too low and is hitting bottom of sample holder or plate in XY)
- b) Clogged sample needle
- c) Syringe plunger/glass assembly not tightened enough into valve
- d) Fitting on Syringe pump not tight
- e) 5 or 50” line not sealing well (check for misshapen or damaged flange on the ends of the tubing)
- f) Sample valve or fuse
- g) Syringe sheath intake line has bubbles (this line is on the left side of the syringe valve)

38) Green laser (can’t get cv’s down after adjusting all lenses):

- a) Check the green laser “Dot” on the PMT w/the dye to ensure that it’s correct in size & shape
- b) “Full Turns,” in or out, may be necessary on the 15mm lens to re-adjust the focal point, especially after replacing laser. **Although a bit harder to tell, this can also be seen when verifying step “a” in that the dot may not look sharply focused
- c) Laser (beam pattern may be irregular)
- d) Laser output power is low
- e) Verify that the signal on the o-scope doesn’t have a large shoulder; it should be uniform

39) Unit won’t calibrate:

- a) Clogged sample needle (verify that the unit takes up the sample)
- b) Clogged injection needle and/or cuvette (may be visible if you run a sample with the dye)
- c) Verify “Target” values for Calibrators and/or Controls
- d) Try second bottle of calibrator or control beads
- e) Make sure the DD(classification) or RP1(reporter) A.P.D. Gain value is high enough to enable the software to see events/reporter signal
- f) Check for drip in waste container (sheath/backflush valve or fuse. verify purges during wash)
- g) Verify “High Voltage” is on and correct
- h) Verify that the “Laser Split” (green offset $\mu\text{sec.}$) is correct; especially after replacing either laser (25 – 50 μs)
- i) Check A/D offsets
- j) Laser(s) not on

40) Unit sheath pressure fluctuates outside of 5% “Green Window” during sample run:

- a) Verify that the procedure for installing the SD was followed (current procedure requires that the air regulator on the LX100 be turned completely clockwise (to the right) prior to installing the SD)
- b) Air pressure regulator (Analyzer or SD)
- c) For analyzers with firmware version 2.3.9 or higher that have a data cable connected to the SD; the analyzer should not fill at all during data acquisition. Verify that the SD firmware is up to date (2.0.5 at minimum)
- d) Air cylinder
- e) Debubbler valve or fuse
- f) Leaking sheath valve

41) Red laser (can't get DD cv's down after adjusting all lenses):

- a) "Full Turns," in or out, may be necessary on the red laser housing to re-adjust the focal point; especially after replacing the red laser
- b) Laser (beam pattern may be irregular)
- c) Laser output power is low
- d) Verify that the signal on the o-scope doesn't have a large shoulder; it should be uniform

42) The 100 bead map displays a shift (to the left or to the right):

- a) Although the entire map is considered to be "shifted", the shift will primarily be seen on the left two columns of the bead map; this is due to the logarithmic scaling. An adjustment of the power of the red laser itself may help to correct this (**Note: Only make very minute adjustments as irreparable damage will occur to the red laser if turned too much**). To move the population of the beads to the right, turn **down** the power, to move the population of the beads to the left, turn **up** the power. Use the potentiometer on the little circuit board of the red laser itself to change the power. After making the adjustment, recalibrate the red laser and then run a new sample and view the map once again. Readjust if necessary, but **remember to keep the power within the range allowed for the red laser (7.5 – 9.5mW)**

43) Bead Map "Smear" or severe scattering of regions:

- a) Digital board
- b) Analog board
- c) End of waste line is submerged causing back pressure
- d) Waste bottle full or cap not vented
- e) Backflush valve

44) Drain Function Not Working (LX100):

- a) On some very early revisions of the Digital board, everything appears to work except the "Drain" function. On some of these early board revisions, a minor modification to the information in the analyzer "Rev" box may be needed. To correct this, open up the maintenance software and highlight the area inside the "Rev" box; delete any information in it (Do Not "Write" This to the New Board Yet). Now place a capitol letter "A" in the box and push down on the space bar three times. Now write this to the board; and after it's complete, restart the analyzer and retry

45) Miscellaneous:

- b) Whenever you upgrade the SD firmware to 2.0.5, you will need to restart the SD; otherwise you may run into some new problems such as the analyzer not starting up properly or no connectivity to the software.
- c) When performing a prime on analyzers with firmware version 2.3.9, there will be an initial purge of fluid to waste before it slows to the commonly know drip; applicable to all other firmware versions
- d) 47 Plex regions are as follows: 1-8, 10, 11, 14, 16, 17, 19, 22-25, 28, 31, 32, 35, 38, 40, 41, 43, 46, 49, 50, 53, 57, 60, 61, 65, 69, 71, 73, 75, 78, 81, 84, 87, 90, 92, 95, 98 & 100
- e) Beads are injected into the cuvette at a rate of 60µl/minute
- f) XY drive mechanism uses one of two types of travel rods for the plate movement (Old – gear, New – Screw)

46) LX100/200 Power Supply Adjustment:

- g) Turn off the instrument.
- h) Disconnect the green laser control box from the Laser Driver Board before performing this procedure.
- i) Turn the instrument back on and check power measurements.
 - i) Coherent power supply:
 - (1) For systems using a Coherent laser and a Resonant Power Technology power supply, the 5 Volt line of the power supply needs to be set between 5.44 and 5.47 Volts. This adjustment must be made before the laser controller is connected to the laser driver board. Place the voltage probes in the 12-pin connector at the left of the power supply, making contact with the terminals for the red and black wires. Adjust the gray pot that is to the right of and above the 12-pin connector on the supply until the voltage is in range (5.44 to 5.47 Volts)
 - (2) Turn the instrument off and reconnect the control box cable previously removed
 - ii) Uniphase power supply:
 - (1) For systems using a Uniphase laser and a Condor power supply, the 5 Volt line of the power supply needs to be set between 5.50 and 5.55 Volts. This adjustment must be made before the laser controller is connected to the laser driver board. Place the voltage probes in the 9-pin connector of the power supply, making contact with the terminals for the red and black wires. Adjust pot R43 (near the 9-pin connector) on the supply until the voltage is in range (5.50 to 5.55 Volts)
 - (2) Turn the instrument off and reconnect the control box cable previously removed

*****If at all uncertain about the flow of a sample through the unit; you can run Laser Alignment Solution (a.k.a. Rhodamine), as a sample and watch the intake from the sample needle, into the sample loop, through the cuvette and through the backflush valve to the waste container.**

Shortcuts, Passwords & More:

1.7

- a) To gain access to the "Simplex" mode: (*Note: Access to this mode will be reset upon program shutdown)
 - i) Open software
 - ii) Click "Options" at the top of the screen
 - iii) Select & hold down the "control", "alt", "M" keys of the keyboard
 - iv) When prompted, type in the password (microsphere) and select "OK" (you should see the "Maint" tab show up on the "Options" screen
 - v) Click "Apply", then "OK" (the "Simplex" icon will be displayed)
- b) To gain access to the "Advanced" mode: (*Note: This mode will remain active, even upon program shutdown)
 - i) Open software
 - ii) Click "options" at the top of the screen
 - iii) Select & hold down the "control", "alt", "A" keys of the keyboard
 - iv) If prompted, type in the password (microsphere) and select "OK" (you should see the "Advanced" tab show up on the "Options" screen) **Note: To ensure that all features are viewable to the engineer, on the "Advanced" tab, everything should be selected with the exception of "Enable Liquid Handler"
 - v) Click "Apply", then "OK"

- c) To gain access to the “File” mode: (*Note: This mode will remain active, even upon program shutdown)
 - i) Open the software (do not select “New Session” or “Simplex” mode at this time)
 - ii) Go into “Options” and make sure that “Allow File Mode” is checked (this can be found on the “Advanced” tab)
 - iii) Place the mouse cursor over the word “Standby” in the upper right hand corner of the software program
 - iv) Hold down the “Ctrl”, “Alt” keys on the keyboard and left click the mouse (while the mouse cursor is over “Standby”)
 - v) Perform the previous two steps again to exit “File” mode
-

IS

- a) Cancelling Warm Up
 - i) Open Luminex IS Software
 - ii) Click the “Run Batch” tab
 - iii) Place the mouse cursor inside the frame surrounding the command buttons (Start, cancel cmd, resume, etc.)
 - iv) Simultaneously hold down the Ctrl + Alt keys and left click the mouse
- b) Displaying the “Advanced” tab (option 1)
 - i) Make a copy of the Luminex IS shortcut found on the Windows desktop
 - ii) Right click the new shortcut and select “Properties”
 - iii) Click on the “Shortcut” tab at the top of the new display
 - iv) Click in the “Target” box where the shortcut is displayed (this area will generally be highlighted already when the properties page opens up)
 - v) Add “ –a” (space, dash, a) to the existing Target text (should be something like:
“C:\ProgramFiles\Luminex\Luminex 100 IS\Lx100IS.exe”
 - vi) Open Luminex IS software and the “Advanced” tab should now be visible.

**Note: The advanced tab is designed for users with advanced or specific knowledge of LXR SDK

- c) Displaying the “Acq. Detail” tab (option 1)
 - i) Make a copy of the Luminex IS shortcut found on the Windows desktop
 - ii) Right click the new shortcut and select “Properties”
 - iii) Click on the “Shortcut” tab at the top of the new display
 - iv) Click in the “Target” box where the shortcut is displayed (this area will generally be highlighted already when the properties page opens up)
 - v) Add “ –lds” (space, dash, lds) to the existing Target text (should be something like:
“C:\ProgramFiles\Luminex\Luminex 100 IS\Lx100IS.exe”
 - vi) Open Luminex IS software and the “Acq. Detail” tab should now be visible.
- d) Option #2 for “Advanced” or “Acq. Detail” tab
 - i) Right click the existing IS shortcut found on the Windows desktop and select “Properties”
 - ii) Click on the “Shortcut” tab at the top of the new display
 - iii) Click in the “Target” box where the shortcut is displayed (this area will generally be highlighted already when the properties page opens up)
 - iv) Add the appropriate extension needed “ –a” or “ –lds” to the end of the existing target
 - v) Click “Apply”, then “OK”
 - vi) Open Luminex IS software and the new tab should now be visible

***Upon completion of services, the copy of the IS icon on the desktop must be deleted or the information added to the original icon target must be removed.

LXR Maintenance (2.0.652 & 2.0.682 SP1)

- a) To gain "Limited" access to advanced settings: (*Note: Access to this mode will be reset upon program shutdown)
 - i) Open software
 - ii) Go to "Service" (top left of screen) and select "Access Level"
 - iii) Type in the password when prompted (y.....b.....b....t....)
 - iv) Select "OK"

- b) To gain unrestricted access (Private Mode): (*Note: Access to this mode will be reset upon program shutdown)
 - i) Open software
 - ii) Complete the steps for "Limited" access
 - iii) Select & hold down the "control", "shift", "P" keys of the keyboard (you should see the letter "P" show up at the bottom of the screen next to the "XYP State" box)

- c) To gain access to the "Manufacturing" tab for the XY: (*Note: Access to this mode will be reset upon program shutdown)
 - i) Open software
 - ii) Complete the steps for "Limited" access
 - iii) Select & hold down the "control", "shift", "M" keys of the keyboard (you should see the letter "M" show up at the bottom of the screen next to the "System State" box)

- d) To modify the Maintenance program so that no access password is required upon star up: (*Note: This is **not** to be done on a customer's pc as this setting is permanent and does not reset when the software is shut down. This only covers the initial password entry; "Ctrl, Shift, P" will still have to be done manually)
 - i) Right click on the Maintenance icon shortcut and select "properties"
 - ii) Locate the "Target" box and add /p:youbetterbettrained to the very end of the existing target (Note: Put a space between the existing target and the new one)
 - iii) Click "Apply", then "OK"

LXR Maintenance

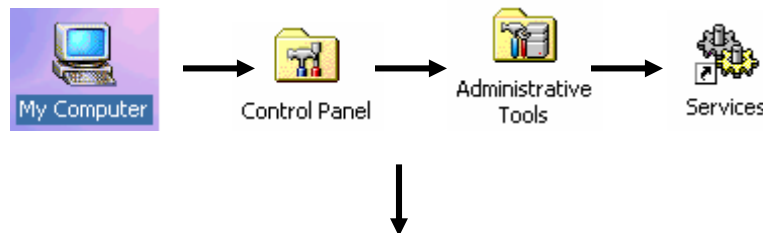
(Changing LXR Services from Auto to Manual)



This symbol tells you that the maintenance services are running. It can be seen at the bottom right of the screen next to the time.

Follow the steps below to change the service so that it will only begin to run when the software is opened.

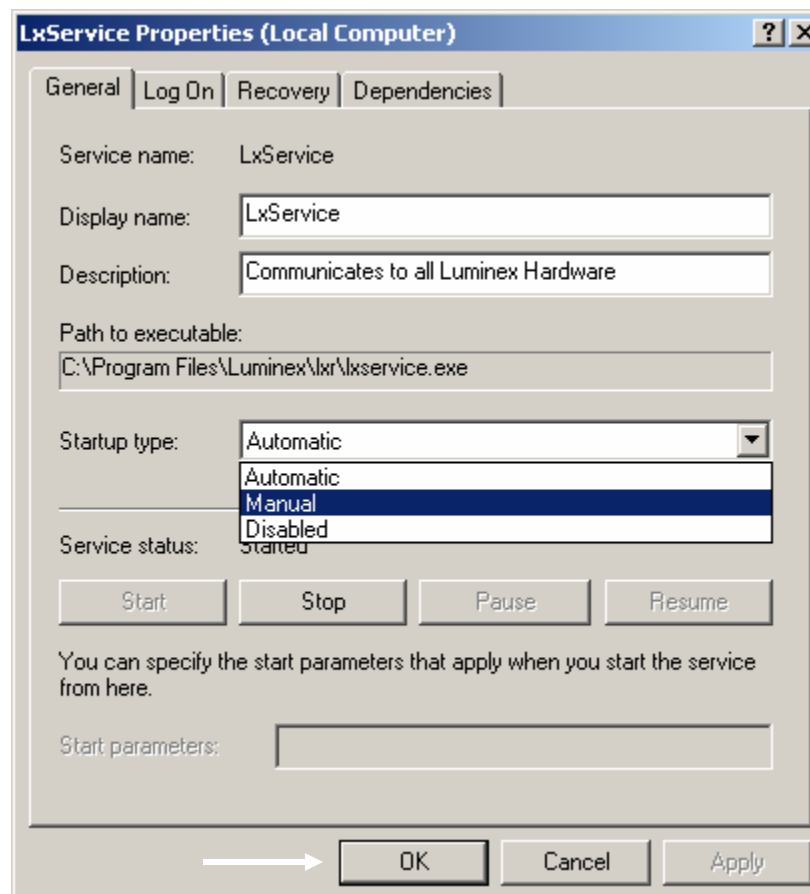
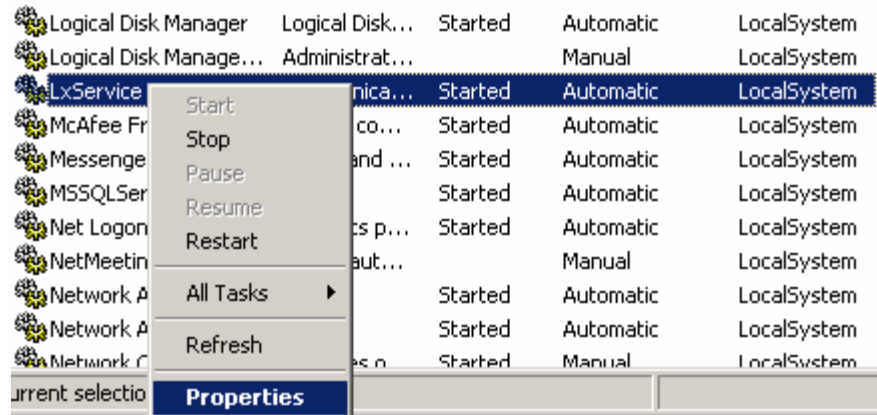
**** This it to be done on your personal laptop or pc only!**



Tree	Name	Description	Status	Startup Type	Log On As
Services (Local)	Alerter	Notifies sel...		Manual	LocalSystem
	Application Manage...	Provides s...		Manual	LocalSystem
	Automatic Updates	Enables th...	Started	Automatic	LocalSystem
	Background Intellig...	Transfers f...	Started	Manual	LocalSystem
	ClipBook	Supports C...		Manual	LocalSystem
	COM+ Event System	Provides a...	Started	Manual	LocalSystem
	Computer Browser	Maintains a...	Started	Automatic	LocalSystem
	DHCP Client	Manages n...	Started	Automatic	LocalSystem
	Distributed Link Tra...	Sends notif...	Started	Automatic	LocalSystem
	Distributed Transac...	Coordinate...		Manual	LocalSystem
	DNS Client	Resolves a...	Started	Automatic	LocalSystem
	Event Log	Logs event...	Started	Automatic	LocalSystem
	Fax Service	Helps you ...		Manual	LocalSystem
	Indexing Service			Manual	LocalSystem
	Internet Connectio...	Provides n...		Manual	LocalSystem
	IPSEC Policy Agent	Manages I...	Started	Automatic	LocalSystem
	Logical Disk Manager	Logical Disk...	Started	Automatic	LocalSystem
	Logical Disk Manage...	Administrat...		Manual	LocalSystem
	LxService	Communica...	Started	Automatic	LocalSystem
	McAfee Framework ...	Shared co...	Started	Automatic	LocalSystem

LXR Maintenance

(Changing LXR Services from Auto to Manual)



RESETTING COM PORT NUMBER ASSIGNMENTS

How can I reset the COM port assignments on my Windows computer? (Windows 2000 & XP).

RESOLUTION

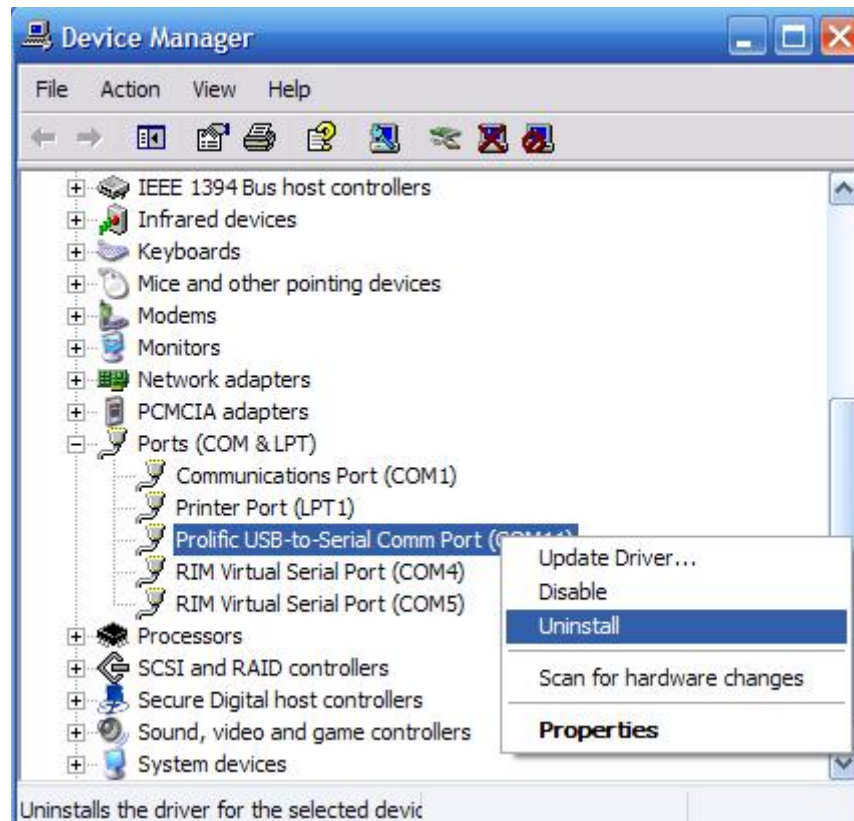
Each time a USB device is plugged into a USB port, Windows "rediscovers" the device. This can lead to having many different COM port assignments, and cause hardware compatibility issues. Resetting the COM database may resolve these issues.

After repeated installations of various USB-serial converters, the assigned com ports continue to increase with the operating system reserving a COM number for every USB device that was ever installed. Even though it is possible to reassign a com number in the Device Manager, some vestige of the old assignments remain and may be polled as part of COM interaction with the application and the operating system.

CAUTION: This procedure requires editing the registry. Please follow the instructions carefully.

1. Uninstall any USB-serial converter drivers or other com port add-in PC cards or PCI cards in the Device Manager. Do this by right clicking on the device in the Ports branch of Device Manager. If you have any kind of multiport serial adapters, uninstall those as well. After they are uninstalled, unplug the USB-serial converter(s) (**See pics below**).

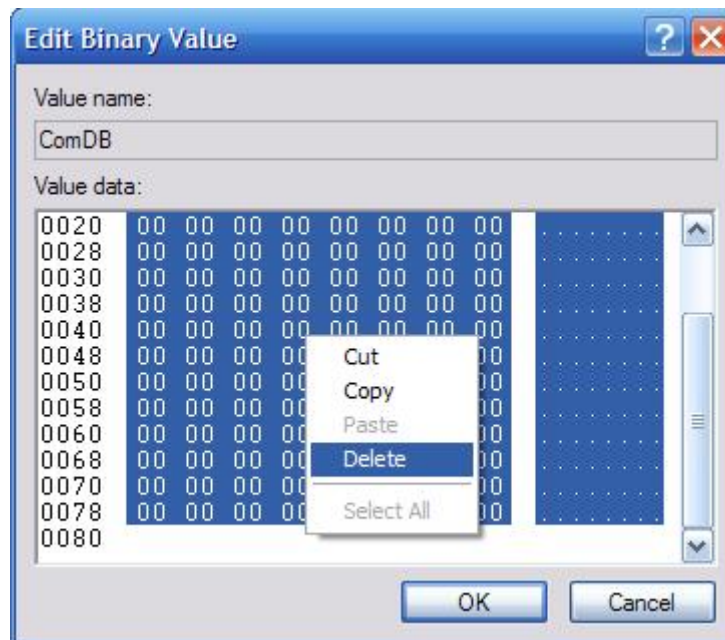
To access the device manager from the desktop: Right click on "My Computer" icon, select properties, hardware, and device manager.



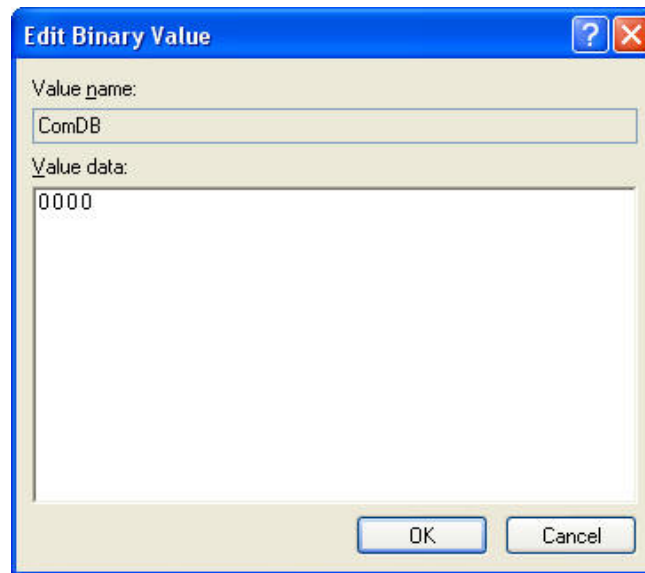


2. From the desktop, Click **Start | Run**, type **RegEdit** and then click **OK**.
3. Navigate to the **KEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\COM Name Arbiter**.
4. Highlight **COM Name Arbiter**.
5. Select **File | Export**.
6. Pick a suitable name and location to save this file. Note where you put it. In case of problems, you can restore exactly where you were with this branch by running **RegEdit** and clicking **File | Import** to import this file.
7. Right-click on the **ComDB** entry on the right side of the window, and then click **Modify** (*!!Do not select delete*).
8. In the Edit Binary Data window, highlight all the values and delete them; this sets them to 0 (**See pics below**).
9. Click "OK"

(Before)



(After)



1. Click **File, Exit** to exit the Registry Editor
2. Restart the computer.
3. All of your native COM ports will be discovered automatically.
4. Any serial-USB converters will need to be reinstalled but they should start with the first COM number after your native COM ports.

NOTE: This procedure is being done from the factory set-up. To accomplish this, perform the following steps:
1: select "Save/Recall" button, 2: At the display select "Recall Factory Setup", 3: Select "OK Confirm Factory Init.", procedure complete.

VERTICAL COLUMN

1. Select "Menu" button
2. At the display select "AC" for "CH1 Coupling & Impedance" & adjust Vertical "Scale" knob to 500mV/div. for "Fine Scale" setting.
3. Use the "position" knob and bring the yellow horizontal signal 1 or 2 grid squares from the bottom of the display.
4. Select CH2 button
5. At the display select "AC" for "CH2 Coupling & Impedance" & adjust Vertical "Scale" knob to 500mV/div. for "Fine Scale" setting.
6. Use the "position" knob and bring blue horizontal signal just below channel 1 signal.

HORIZONTAL COLUMN

1. Select "Delay" button to turn "Off" the function.
2. Adjust "Scale" knob to show "10 μ s" (microseconds) at the bottom of display screen.

TRIGGER COLUMN

1. Select "Trigger" button, then select the desired trigger channel; in this case we'll start with "A". (Triggering can be done from either channel. However, it is recommended that the trigger be set to the channel that is not being adjusted. Doing so will ensure that there is always one steady signal on the scope in the event that the signal for the channel being adjusted is lost).
2. Use level knob and place the trigger level, little arrow on the right side of display, so that it is approximately half a grid square above the channel 1 signal.

ACQUIRE COLUMN

1. Select "Menu" button
2. At the display select "Average" button
By default the Average will be 16. You may need to adjust this average, using the single knob at the top of the scope, (one increment at a time) to help keep your signal steady or to help the response time of your signal on the scope as you're making adjustments to the lasers. **Note:** as you increase this number, the response time on the o-scope to adjustments made will also increase.

TOP FUNCTION BUTTONS

1. Select "Cursor" button
2. At the display select "V Bars" button
Using Upper knob and Select button (to toggle between the two bars), you can adjust the bars for your signals offset between DD and RP1. The offset will be seen at the top of the screen displayed next to the "Delta" symbol on the left.
3. Select "Save/Recall" button at the display select "Save Current Setup". At the display select "To Setup1". This will save all of your current settings and allow you to recall this setup should there be any future confusion in the settings. To recall the setup you saved, select "Save/Recall" button at the display select "Recall Saved Setup". At the display select "Setup 1"; then you're there. **Note:** You can use any of the 3 "Setup" buttons to save and recall your current settings.
4. The Quick menu button will allow you to see your overall set-up.

NOTE: Remember that the RP1 signal should ALWAYS be to right of the Classification signal.

NOTE: This procedure is being done from the factory set-up. To accomplish this, perform the following steps:
1: close the blue screen (if applicable), 2: select "User" button, 3: "Options", 4: use arrow keys and select "Factory Default" and "Enter", 5: select "Yes", procedure complete. **Factory Default will erase all saved setups.

*Use the arrow keys to navigate through the screens. When using the "Range" and "Move" buttons, be sure that you are adjusting the correct channel as there are identical buttons for both the A & B channel.

"A" CHANNEL

1. Select "A" button
2. At the display select AC for "Coupling".
3. Adjust "Range" button for 500mV at bottom left of screen.
4. Use "Move" button and place channel "A" reference line closer to bottom of screen.

"B" CHANNEL

5. Select "B" button
6. At the display select AC for "Coupling".
7. Select "On" for "Input B".
8. Adjust "Range" button for 500mV at bottom right of screen.
9. Use "Move" button and place channel "B" reference line on or just below "A" reference line.

Waveform Spec's

10. Select "Scope" button
11. At the display select "On" for "Readings".
12. Select "Waveform Options".
13. Use arrow keys to navigate to "Average" and select "On" and "Enter".
14. Select "Average 8 or 64" and "Enter" (The difference between the two is that the lower number will cause the signal to react quicker as you make optical adjustments, but the signal may not be very steady. The higher number will cause the signal to react more slowly as you make optical adjustments, but may help to steady the signal better than the lower number).
15. Using the "Time" button, adjust the setting at the bottom of the display to read 10µs (microseconds).

TRIGGER

16. Select "Trigger" button, then select the desired trigger channel; in this case we'll start with "A". (Triggering can be done from either channel. However, it is recommended that the trigger be set to the channel that is not being adjusted. Doing so will ensure that there is always one steady signal on the scope in the event that the signal for the channel being adjusted is lost).
17. Use "Arrow" button to bring trigger level indicator to approximately 1 grid square above the "A" reference line.
18. Use left right "Arrow" button to adjust trigger level indicator so that it is centered over the "Auto Level" box in the display.

V-Bars

19. Select "Cursor" button
20. Using the up or down "Arrow" buttons to select between the two bars, bring them to the center of the screen and adjust them so that they are 35µs (micro seconds) apart. The right reading on the red display at the top of the screen will indicate the distance.

SAVE/RECALL

21. Select "Save/Print" button (This will cause the scope to go into "Hold" as seen on display)
22. Select "Save".
23. Select a number to save your settings at. (By default, #1 will be selected to save the "Screen + Setup", but you can select any one of the locations in this section).
24. Select "Save" at the display.
25. Select "Clear Menu".
26. Press "Hold/Run" button to release "Hold" (Display should show "1/2 Auto in upper right corner")

27. To Recall a setting select "Save Print" button
28. Select "Recall".
29. Use "Arrow" buttons to select number of saved setup.
30. Select "Recall Setup".

CAUTION: If the "Auto/Manual" button is pressed any time during use, you will loose your settings and will have to recall your set up using steps 27-30.

NOTE: Remember that the RP1 signal should ALWAYS be to right of the Classification signal.

Alignment Microspheres

(LXR Maintenance 2.0.682)

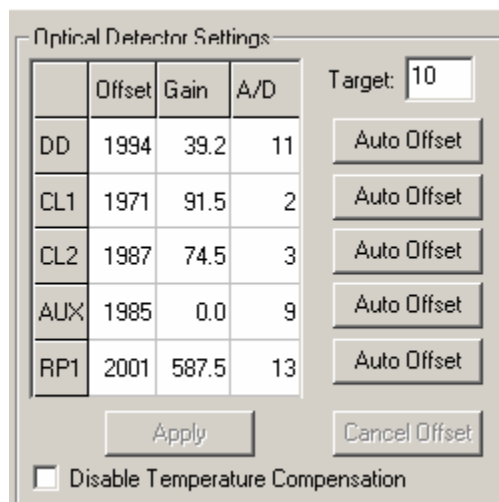
**This procedure is geared towards fine adjustments only, and set up to run using the single sample holder. It will begin as though you have warmed up the system, completed any course alignment, if applicable, and calibrated and/or attempted to calibrate both Cal 1 and Cal 2 with LXR Maintenance.

SETTINGS:

1. In the maintenance software, go to the "Settings" tab
2. Under "Region Map Setup" select "Deactivate All"
3. Leave the "Gate Setup" as is (no gate adjustment is needed for this procedure)
4. Under ""Auto-stop Setup" select "None"
5. Set "Sample Size μ l" to 200
6. Click "Apply" (lower right corner)
7. Select "Detector 1" tab (upper left corner)
8. At the top left corner of software, select "Service", "Access Level"



9. When prompted, enter password, and select "OK" (**NOTE:** there's no prompt to indicate incorrect entry of password. If password is entered correctly, then the functions associated with password entry will work)



10. In the "Optical Detector Settings" section, highlight and change the "Gain" voltage values for CL1 & CL2 to: CL1 (50), CL2 (60)

Alignment Microspheres

(LXR Maintenance 2.0.682)

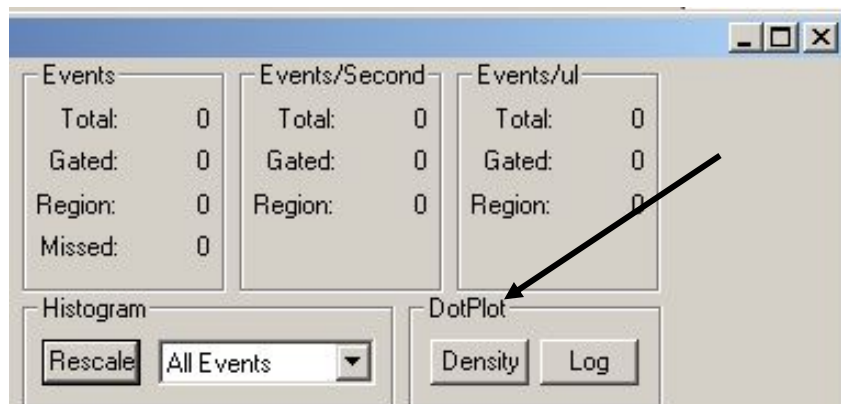
11. Once the values have been changed, select “Apply”, then “OK” to confirm change
(**NOTE:** The alignment microsphere beads are very bright and if run without this adjustment, the classification population will be barely visible on the right edge of the Dot Plot. Selecting apply will change the voltages sent to CL1 & CL2 in the u-block, adjusting the placement of the alignment microspheres on the dot plot. (No “Gain” adjustment is necessary when aligning the reporter channel))
12. Enable “Disable Temperature Compensation” by selecting check box (**NOTE:** This selection will hold the voltages just entered indefinitely as long as the box is checked. Leaving this box checked does not affect calibration, but it must be unchecked prior to disconnecting from unit)
13. The “Settings” Portion is now complete

SIMPLEX DISPLAY:

14. Select “Tools” just above the “Detector 1” tab
15. Under “Tools” select “Simplex Display” (A second screen showing Simplex Display will open up)

NOTE: At this point you will have to toggle between the new “Simplex Display” screen and the “Detector 1” screen. When you go to reselect the “Simplex Display” screen, take care **not** to “click” the mouse anywhere in the “Dot Plot” area of the display. You will add a “Region” to the display without wanting one.

16. With your vortexed Alignment Microsphere beads in the sample holder select “Run Sample” under “Detector 1”
17. Switch back to simplex display
18. Just above the simplex display, click the button that reads “Density” to change it to “Decaying” (Decaying only shows the beads that are currently passing through the cuvette, whereas Density is accumulative and shows every bead that has passed through the cuvette during a sample run)



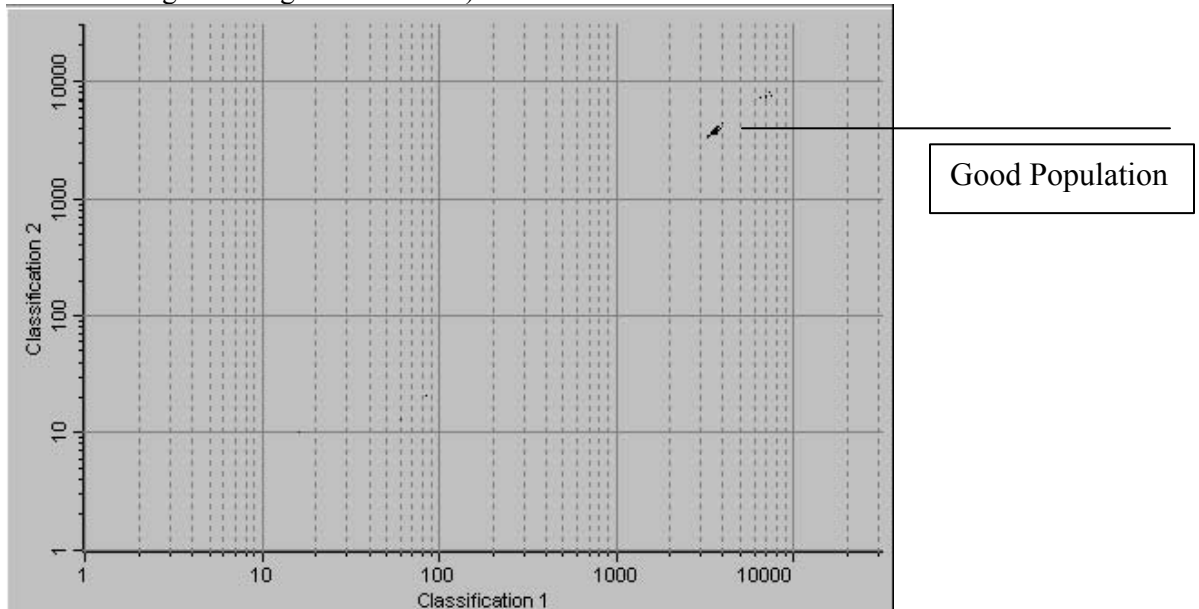
THIS DOCUMENT IS FOR REFERENCE ONLY AND IS INTENDED FOR USE BY
LUMINEX TRAIND FIELD SERVICE PERSONNEL

Updated 5/8/08 D.L.R.

Alignment Microspheres

(LXR Maintenance 2.0.682)

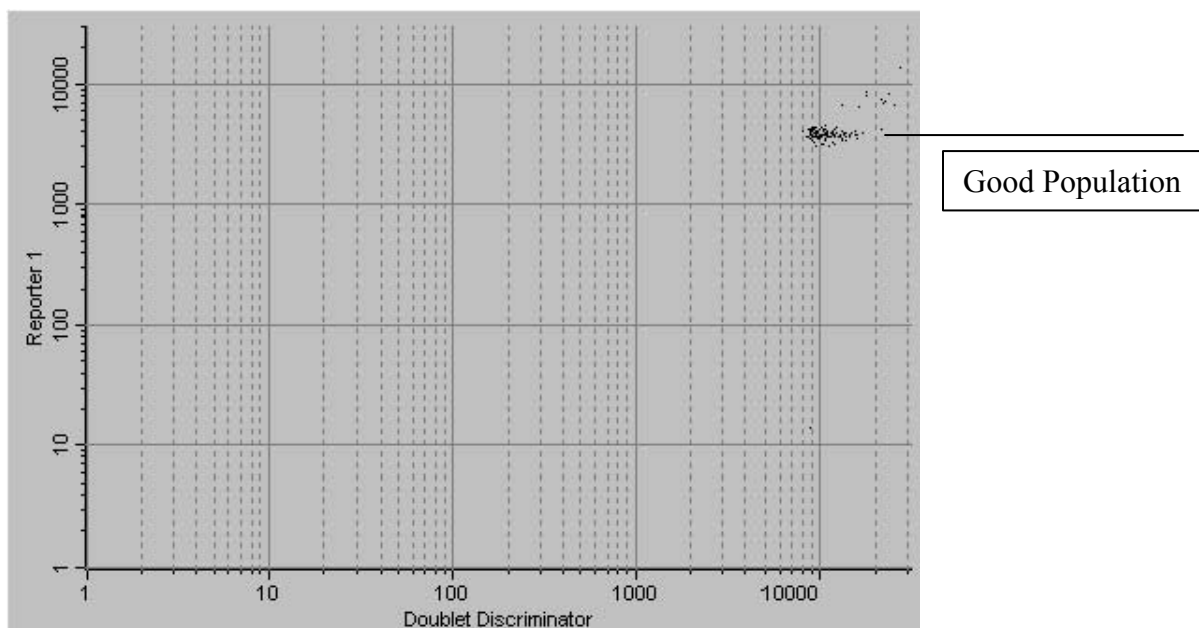
19. Beginning with the red laser housing, make adjustments to each lens screw while watching the dot plot screen. The goal is to make the population as small as possible, which happens when the population moves to the upper right corner of the dot plot (Adjusting the entire classification channel may be necessary to obtain this result. Always adjust each screw on the lens to make the population as good as it can possibly be before moving on to align the next lens)



20. Continue down the optical path and adjust the dichroic mirror, then continue on to CL1 & CL2
21. When finished aligning the classification channel, anywhere in the “Dot Plot” area, right “click” the mouse and select “X” axis, then “Doublet Discriminator”. Right “click” the mouse again and select “Y” axis, then “Reporter”
22. With your vortexed Alignment Microsphere beads in the sample holder select “Run Sample” under “Detector 1”
23. Switch back to simplex display
24. Beginning with the 10/15mm lens on the reporter channel, make adjustments with each lens while watching the dot plot screen. The goal is to make the population as small as possible, which happens when the population moves to the upper right corner of the dot plot (Adjusting the entire reporter channel may be necessary to obtain this result. Always adjust each screw on the lens to make the population as good as it can possibly be before moving on to align the next lens)

Alignment Microspheres

(LXR Maintenance 2.0.682)



25. Continue down the optical path and adjust the dichroic mirror
26. When finished aligning, run calibration for both Classification and Reporter channel
27. Perform “Verification” procedure to obtain new cv and mean results
28. Re Run this procedure if necessary to further improve results
29. When alignment is complete, Un check the box for “Disable Temperature Compensation” if still selected
30. Procedure complete

Calibrators

(LXR Maintenance 2.0.682)

**This procedure is geared towards fine adjustments only, and set up to run using the single sample holder. It will begin as though you have warmed up the system, completed any course alignment, if applicable, and calibrated and/or attempted to calibrate both Cal 1 and Cal 2 with LXR Maintenance.

SETTINGS:

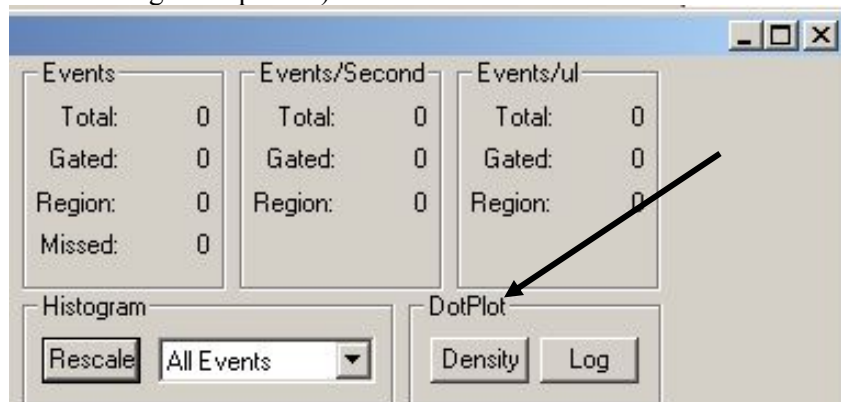
1. In the maintenance software, go to the “Settings” tab
2. Under “Region Map Setup” select “Deactivate All”
3. Leave the “Gate Setup” as is (no gate adjustment is needed for this procedure)
4. Under “Auto-stop Setup” select “None”
5. Set “Sample Size μl ” to 200
6. Click “Apply” (lower right corner)
7. Select “Detector 1” tab (upper left corner)
8. The “Settings” Portion is now complete

SIMPLEX DISPLAY:

9. Select “Tools” just above the “Detector 1” tab
10. Under “Tools” select “Simplex Display” (A second screen showing Simplex Display will open up)

NOTE: At this point you will have to toggle between the new “Simplex Display” screen and the “Detector 1” screen. When you go to reselect the “Simplex Display” screen, take care **not** to “click” the mouse anywhere in the “Dot Plot” area of the display. You will add a “Region” to the display without wanting one.

11. With your vortexed Calibrator 1 beads in the sample holder select “Run Sample” under “Detector 1”
12. Switch back to simplex display
13. Just above the simplex display, click the button that reads “Density” to change it to “Decaying” (Decaying only shows the beads that are currently passing through the cuvette, whereas Density is accumulative and shows every bead that has passed through the cuvette during a sample run)



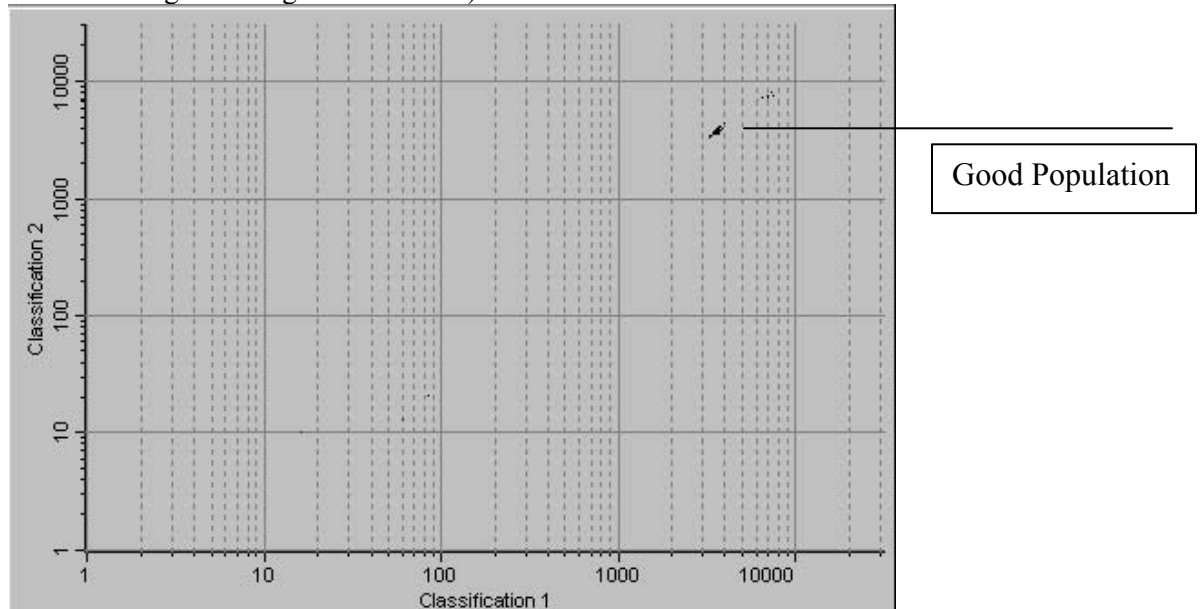
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Updated 5/8/08 D.L.R.

Calibrators

(LXR Maintenance 2.0.682)

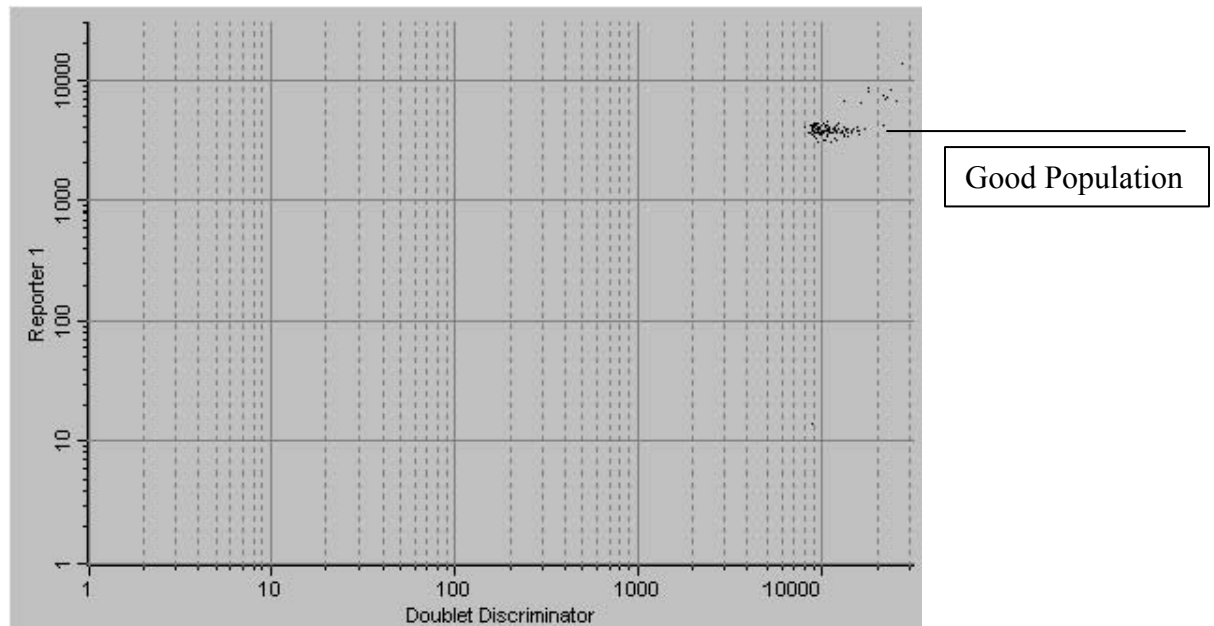
14. Beginning with the red laser housing, make adjustments to each lens screw while watching the dot plot screen. The goal is to make the population as small as possible, which happens when the population moves to the upper right corner of the dot plot (Adjusting the entire classification channel may be necessary to obtain this result. Always adjust each screw on the lens to make the population as good as it can possibly be before moving on to align the next lens)



15. Continue down the optical path and adjust the dichroic mirror, then continue on to CL1 & CL2
16. When finished aligning the classification channel, anywhere in the “Dot Plot” area, right “click” the mouse and select “X” axis, then “Doublet Discriminator”. Right “click” the mouse again and select “Y” axis, then “Reporter”
17. With your vortexed Calibrator 2 beads in the sample holder select “Run Sample” under “Detector 1”
18. Switch back to simplex display
19. Beginning with the 10/15mm lens on the reporter channel, make adjustments with each lens while watching the dot plot screen. The goal is to make the population as small as possible, which happens when the population moves to the upper right corner of the dot plot (Adjusting the entire reporter channel may be necessary to obtain this result. Always adjust each screw on the lens to make the population as good as it can possibly be before moving on to align the next lens)

Calibrators

(LXR Maintenance 2.0.682)



20. Continue down the optical path and adjust the dichroic mirror
21. When finished aligning, run calibration for both Classification and Reporter channel
22. Perform “Verification” procedure to obtain new cv and mean results
23. Re Run this procedure if necessary to further improve results
24. Procedure complete

Field Service

LXR 2.0.652 & 2.0.682 Verification Procedure

****This procedure will begin as though you have already connected, warmed up the system and successfully calibrated Cal 1 and Cal 2 with LXR Maintenance.**

SETTINGS:

1. Go to the “Settings” tab
2. Under “Region Map Setup” select “Deactivate All”
3. Under “Gate Setup” you can now set your gate values if you so choose to (it is also possible to do it later in the procedure as well)
****Remember that your nominal gate values are 8,500 – 13,500. But these values are based on a DD target value of 10,000. Adjust your gate settings accordingly, for the difference, if your DD target value is higher or lower.**
4. Under “Auto-stop Setup” select “Total Count”
5. Set “Count” Max value to 2000 (events)
6. “Sample Size μ l” is correct by default (50)
7. Click “Apply” at bottom of screen
8. The “Settings” Portion is now complete

SIMPLEX DISPLAY:

9. Select “Detector 1” tab at the top of screen.
10. Select “Tools” just above the “Detector 1” tab
11. Under “Tools” select “Simplex Display” (Simplex Display screen will now be seen over LXR Maintenance)

****At this point you will have to toggle between the new “Simplex Display” screen and the “Detector 1” screen. When you go to reselect the “Simplex Display” screen, DO NOT “click” the mouse anywhere in the “Dot Plot” area of the display. You will add a “Region” to the display without wanting one.**

12. With your vortexed Cal 1 beads in the sample holder select “Run Sample” under “Detector 1”
13. When the sample is finished select “Rescale” in the “Simplex Display”
14. If you have already pre selected your gates, continue on to #15. If not, then “drag” each gate to the position needed by placing the mouse cursor over the gate, then left “click” the mouse and move it. You can see the gate values in the lower left corner of the screen.
15. Now Reselect “Gated Events” in the “Histogram” section (even though it may already show “Gated Events”, it will not actually gate anything until reselected)
16. Create a “Region” over the population of collected beads in the “Dot Plot” area. To do this, move the mouse cursor anywhere in the “Dot Plot” area, right “click” the mouse, select “Region”, and then select “Create” (the mouse cursor will change to a “+”). Place the “+” symbol in the center of the population, left “click” the mouse and create the region. If the population is larger than the region, left click the mouse again and make the region larger to cover the entire population. ****We are primarily concerned with the main population and not the few individual beads that may be located farther away.**
17. Now select “Run Sample” again and run the Cal 1 beads a second time

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Field Service

LXR 2.0.652 & 2.0.682 Verification Procedure

18. When the sample is completed, select the needed result ("%CV", "Mean" or "Peak") on the "Simplex Display" next to "Statistic" (near the top left of the screen). The results will now be displayed.
**Now the X & Y axis must be changed to view the population of the Reporter laser when running Cal 2.
19. Anywhere in the "Dot Plot" area, right "click" the mouse and select "Doublet Discriminator" for "X" axis and "Reporter" for the "Y" axis
20. With your vortexed Cal 2 beads in the sample holder select "Run Sample" under "Detector 1"
21. Create a "Region" over the population of collected beads in the "Dot Plot" area. To do this, move the mouse cursor anywhere in the "Dot Plot" area, right "click" the mouse, select "Region", and then select "Create" (the mouse cursor will change to a "+"). Place the "+" symbol in the center of the population, left "click" the mouse and create the region. If the population is larger than the region, left click the mouse again and make the region larger to cover the entire population. **We are primarily concerned with the main population and not the few individual beads that may be located farther away.
22. Now select "Run Sample" again and run the Cal 2 beads a second time
23. When the sample is completed, select the needed result ("%CV", "Mean" or "Peak") on the "Simplex Display" next to "Statistic" (near the top left of the screen). The results will now be displayed.
24. **Record the results before closing the "Simplex Display" screen as all information will be lost once it is closed.**

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Update 5/26/04

D.L.R.

(BREAKDOWN)
LOT #A6428 or EQUIVALENT

System Control Trend Report

7/29/2008 11:30:00 AM

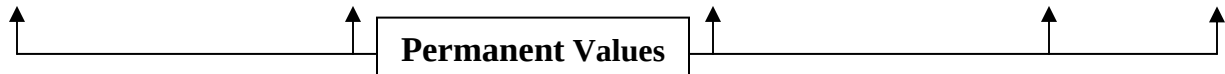
Product Number: L100-CON1
Product Name: L100-CON1

Report Start Date: XXXX
Report End Date: XXXX

Lot Number: *A6428
Expiration Date: XXXX

•Target Information•

Region	Count (Min)	A1 (Target Values)	A2 (Max CV)	B1 (Target Values)	B2 (Max CV)	C1 (Target Values)	C2 (Max CV)	D1 (Max)
1	1000	*9917	20	*694	8.5	*207	9.6	100
2	1000	*9928	20	*68.0	9.49	*65.7	9.81	100
3	1000	*10717	20	*23112	7.55	*5580	9.12	100
4	1000	*9542	20	*2426	7.66	*17632	8.56	100
5	1000	*9700	20	*217	8.68	*5728	9.11	100

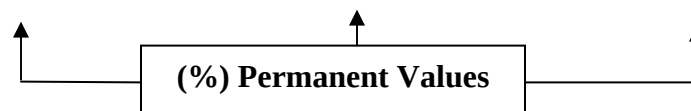


•Region Results•

Time	Total Beads	Total Gated Beads	Pass/Fail	Lot Expired	Serial Number
XXXX	XXXX	XXXX	P/F	Yes/No	LX100...

•Detail Results•

Region (Actual)	Count (Min. 1000)	A1 DD Peak (+/- % Of Target)	A2 DD CV	B1 CL1 Mean (+/- % Of Target)	B2 CL1 CV	C1 CL2 Mean (+/- % Of Target)	C2 CL2 CV	D1 RP1 Mean
21	X	(15.5%) *8380-11454	X	(8.3%) *636-752	X	(8.0%) *190-224	X	X
6	X	(13.9%) *8548-11308	X	(35.5%) *44-92	X	(8.1%) *60.38-71.02	X	X
81	X	(18.8%) *8702-12732	X	(9.8%) *20847-25377	X	(10.6%) *4989-6171	X	X
96	X	(15.9%) *8025-11059	X	(8.6%) *2217-2635	X	(6.4%) *16504-18760	X	X
73	X	(15.4%) *8206-11194	X	(12.5%) *190-244	X	(5.9%) *5390-6066	X	X



*Range values are lot specific. Those listed are valid for lot **A6428** or any lot with the same target values.

THIS DOCUMENT IS FOR REFERENCE ONLY

UPDATED 7/29/2008

DLR/MEH



CON.1 TREND REPORT

(BREAKDOWN)

LOT #A5272 or EQUIVALENT

System Control Trend Report

10/17/2006 12:00:00 PM

Product Number: L100-CON1

Report Start Date: XXXX

Product Name: L100-CON1

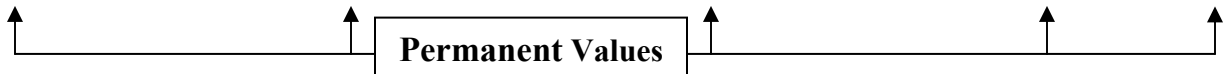
Report End Date: XXXX

Lot Number: ***A5272**

Expiration Date: XXXX

•Target Information•

Region	Count (Min)	A1 (Target Values)	A2 (Max CV)	B1 (Target Values)	B2 (Max CV)	C1 (Target Values)	C2 (Max CV)	D1 (Max)
1	1000	*10017	20	*723	8.5	*201	9.6	100
2	1000	*10047	20	*66.1	9.49	*67.6	9.81	100
3	1000	*11098	20	*22834	7.55	*5412	9.12	100
4	1000	*9071	20	*2358	7.66	*17622	8.56	100
5	1000	*9713	20	*227	8.68	*5753	9.11	100

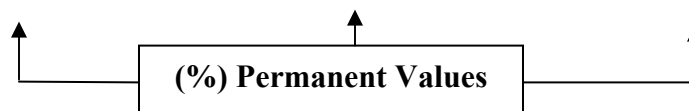


•Region Results•

Time	Total Beads	Total Gated Beads	Pass/Fail	Lot Expired	Serial Number
XXXX	XXXX	XXXX	P/F	Yes/No	LX100...

•Detail Results•

Region (Actual)	Count (Min. 1000)	A1 DD Peak (+/- % Of Target)	A2 DD CV	B1 CL1 Mean (+/- % Of Target)	B2 CL1 CV	C1 CL2 Mean (+/- % Of Target)	C2 CL2 CV	D1 RP1 Mean
21	X	(15.5%) *8464 - 11570	X	(8.3%) *663 - 783	X	(8.0%) *185 - 217	X	X
6	X	(13.9%) *8650 - 11444	X	(35.5%) *42.6 - 89.6	X	(8.1%) *62.1 - 73.1	X	X
81	X	(18.8%) *9012 - 13184	X	(9.8%) *20596 - 25072	X	(10.6%) *4838 - 5986	X	X
96	X	(15.9%) *7629 - 10513	X	(8.6%) *2155 - 2561	X	(6.4%) *16494 - 18750	X	X
73	X	(15.4%) *8217 - 11209	X	(12.5%) *199 - 255	X	(5.9%) *5414 - 6092	X	X



*Range values are lot specific. Those listed are valid for lot **A5272** or any lot with the same target values.

THIS DOCUMENT IS FOR REFERENCE ONLY

UPDATED 10/17/06

DLR/GB



CON.2 TREND REPORT

(BREAKDOWN)

LOT #A5243 or EQUIVALENT

System Control Trend Report

10/17/2006 12:00:00 PM

Product Number: L100-CON2

Report Start Date: XXXX

Product Name: L100-CON2

Report End Date: XXXX

Lot Number: *A5243

Expiration Date: XXXX

•Target Information•

Region	Count (Min)	A1 (Target Values)	A2 (Max CV)	
1	1000	*29.4	28	
2	1000	*180	15	
3	1000	*1894	13	
4	1000	*16896	12	

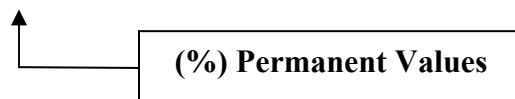


•Region Results•

Time	Total Beads	Total Gated Beads	Pass/Fail	Lot Expired	Serial Number
XXXX	XXXX	XXXX	P/F	Yes/No	LX100...

•Detail Results•

Region	Count (Min. 1000)	A1 RP1 Mean (+/- % Of Target)	A2 RP1 CV	
1	X	(10%) *26.46 – 32.34	X	
2	X	(10%) *168 – 198	X	
3	X	(10%) *1705 – 2083	X	
4	X	(10%) *15206 – 18586	X	



*Range values are lot specific. Those listed are valid for lot A5243 or any lot with the same target values.

THIS DOCUMENT IS FOR REFERENCE ONLY

UPDATED 10/17/06

DLR/GB

TSB100 (LX100)

TSB100-	Priority**	Rev.	Effect Date	Descriptions
001	NS	1	9/23/99	Backflush System Install & Optics Platform Upgrade
002	NF	1	4/17/00	New Laser Install, Brimrose to Uniphase Upgrade
003	NS	1	9/23/99	DSP 3.3 Code Upgrade
004	NS	1	1/20/00	TSB Tracking Label
005	NS	1	9/23/99	DSP 3.4 Code Upgrade
006	NS	1	9/23/99	Remove U54, Install Jumpers & Resistors
007	AR	1	9/23/99	Change Air Flow w/Uniphase Laser
008	AR	1	2/11/99	Pickup Arm Assembly Installation Used w/XYP Tables
009	AR	1	10/2/00	Bio-Rad Resistor Change
010	NS	1	1/10/00	Digital Board Buffer Change
011	NS	1	2/11/99	Larger Screws for V-Housing Assembly
012	NS	1	6/12/00	LX100 Instrument Label UL Requirements Compliance
013	RO	1	6/12/00	Heater Block Storage
014	NS	1	7/7/00	Bubble Trap Retrofit Kit
015	NS	2	Obsolete	DeBubbler Top Upgrade (replaced by TSB100-024)
016	NS	1	7/3/01	Bubble Trap Fittings Upgrade
017	NS	A	4/30/04	Red Laser Power Adjustment
018	NS	1	2/22/02	xMAP Label
019	NS	1	10/9/02	Uniphase Lasers Label
020	AR	B	6/11/04	Red Laser Mount Assy. & 15mm Lens Holder Assy.
021	PO	B	6/15/04	Coherent Laser Retrofit
022	NF	A	1/12/04	LX100 Air Compressor Retro-fit Kit
023	NF	A	9/30/03	Laser Driver Board
024	NS	A	5/31/04	New DeBubbler Top
025	NF	A	5/18/04	PMT Filter
026	NF	A	2/18/05	Lee Co Backflush Valve
027	NF	B	12/13/05	Optimized Analog Board
028	NF	B	11/23/05	New KNF Air Compressor
029	NF	A	2/1/07	Pneumatic Line Check Valve
030	NF	A	7/2/07	XCalibur Syringe Pump
031	NF	A	2/5/10	3-way Sample Valve Change
032	NF	A	3/16/11	Norvell Power Supply
033	NF	A	7/25/11	U8 EPROM Replacement for System Start Up "DSP CRC" Failure

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TSB200 (XYP)

TSB200-	Priority**	Rev.	Effective	Description
(None)				

TSB300 (SDS)

TSB300-	Priority**	Rev.	Effective	Description
001	NS	1	7/3/01	Reservoir Fittings Upgrade

TSB400 (HTS)

TSB400-	Priority**	Rev.	Effective	Description
001	NF	A	5/18/04	PMT Filter
002	NF	A	7/25/11	U8 EPROM Replacement for System Start Up "DSP CRC" Failure

TSB500 (100E)

TSB500-	Priority**	Rev.	Effective	Description
001	NF	B	1/18/06	Lee Co Backflush Valve
002	*	*	*	Not Used
003	NF	A	8/31/07	XCalibur Syringe Pump
004	*	*	*	Not Used
005	NF	A	7/25/11	U8 EPROM Replacement for System Start Up "DSP CRC" Failure

TSB600 (LX200)

TSB600-	Priority**	Rev.	Effective	Description
001	NS	A	5/24/06	Firmware Upgrade To Activate Sample Arm "Up" Position Sensor
002	NF/PO	A	9/25/06	Air Cylinder Regulator Retrofit Kit

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Luminex

Technical Service Bulletins Priority List

Document Number: 00000

Effective: 8/15/11

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Rev. N/A

003	NF	A	2/1/07	Pneumatic Line Check Valve
004	NF	A	7/2/07	XCalibur Syringe Pump
005	NF	A	12/22/09	215M to 115M Coherent Laser Change
006	NF	A	2/5/10	3-way Sample Valve Change
007	NF	A	3/16/11	Norvell Power Supply
008	NF	A	7/25/11	U8 EPROM Replacement for System Start Up "DSP CRC" Failure

TSB700 (FM3D)

TSB700-	Priority**	Rev.	Effective	Description
001	AR	A	7/14/09	Fluidics Bay Front Panel Change
002	NS	A	3/31/10	Laser Controller Board Fuse Change
003	AR	B	4/26/10	Y-Axis Motor Mount Change
004	AR	A	4/26/10	Patch IXPO-0005381
005	AR	C	5/24/10	Modify Sample Agitation Speed
006	AR	A	6/24/09	Reinforced XY Door
007	*	*	*	Not Used
008	*	*	*	Not Used
009	NS	A	1/26/09	New Rotor and Stator Cap
010	NS	A	5/5/11	Probe Coax Coupler Cable

TSB800 (MAGPIX)

TSB800-	Priority**	Rev.	Effective	Description
001	AR	A	6/10/10	Y-axis Upgrade For Assay Prep Module

**** PRIORITY CODES****

I = Immediate

NS = next service call

NF = next failure

AR = as required

RO = request option

PO = purchase option

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LX100/200 Standard Tools

1. Set of US standard BallPoint Allen Hex Wrenches (1/16 through 3/16)
2. Straight Blade Screwdriver (Small Blade and Medium)
3. Phillip Screwdriver (#1, #2)
4. Needle Nose pliers (Small)
5. Bent Nose pliers (Small)
6. Slip Joint Pliers
7. Diagonal Cutters
8. US standard socket set
9. Small Crescent wrench

LX100/200 Specialty Tools

1. Laser Power Meter (Coherent)
 - a. Model: LaserCheck
 - b. Adj. Wavelength 400 – 1064nm
2. Laser Safety Glasses (Laser Shield)
 - a. 190 – 400nm OD5+
 - b. 510 – 670nm OD1.5+
3. Oscilloscope, 2-channel digital
 - a. Tektronix TDS 3032: 300MHz, 2.5GS/s
 - b. Fluke 199C: 200MHz, 2.5GS/s
4. Alignment Microspheres (Luminex)
 - a. FS-0215-01, 10ml
5. Laser Alignment Solution (Luminex)
 - a. FS-0107-01, 10ml
6. CAL-1 & CAL-2 Beads (Luminex)
 - a. L100-CAL1 & L100-CAL2 (5ml bottles)
7. CON-1 & CON-2 Beads (Luminex)
 - a. L100-CON1 & L100-CON2 (5ml bottles)
8. MAG CAL-1 & MAG CON-1 Beads (Luminex)
 - a. MCAL1-05 & MCON1-05 (5ml bottles)
9. Syringe Removal Tool
 - a. FS-0213-01 (old model)
 - b. FS-0297-01 (new model)
10. Cuvette Declogging Wire
 - a. FS-0248-01
11. Loctite #425 (for optical platform)
 - a. Available through Loctite
12. Custom Sample Holder for LX200 (Luminex)
 - a. FS-0253-01
13. Field Service Engineer Kit (Luminex)
 - a. FS-0175-01